Gastrointestinal Physiology

Gastrointestinal Motility

Interdigestive Phase/MMC

1. Which of the following best describes small intestinal motility?

(A) Contractile frequency is constant from duodenum to terminal ileum.
(B) Peristalsis is the only contractile activity that occurs during feeding.
(C) Migrating motor complexes occur during the digestive period.
(D) Vagotomy abolishes contractile activity during the digestive period.
(E) Contractile activity is initiated in response to bowel wall distention.

Answer: E. Contractile activity in the small intestine is initiated in response to distention of the bowel wall. Three types of smooth muscle contractions contribute to small intestinal motility-peristalsis, segmental contractions, and tonic contractions. A fourth type of contraction, peristaltic rushes, are very intense peristaltic waves that may occur in intestinal obstruction. The basal electrical rhythm (BER) are the spontaneous rhythmic fluctuations in membrane potential in the smooth muscle along the GI tract. The BER itself rarely causes muscle contraction, but contractions only occur during the depolarizing phase of BERs, which function to coordinate the various types of contractile activity. The BER is initiated by the interstitial cells of Cajal, which, in the small intestine, are located in the outer circular muscle layer near the myenteric plexus. There are an average of approximately 12 BER cycles/min in the duodenum and proximal jejunum and 8/min in the distal ileum. During fasting between periods of digestion, cycles of motor activity called migrating motor complexes (MMC), migrate from the stomach to the distal ileum. The MMCs immediately stop with ingestion of food. After vagotomy, contractile activity becomes irregular and chaotic.

2. Slow waves in small intestinal smooth muscle cells are

(A) Action potentials
(B) Phasic contractions
(C) Tonic contractions
(D) Oscillating resting membrane potentials
(E) Oscillating release of cholecystokinin (CCK)

Answer: D. Slow waves are oscillating resting membrane potentials of the gastrointestinal (GI) smooth muscle. The slow waves bring the membrane potential toward or to threshold, but are not themselves action potentials. If the membrane potential is brought to threshold by a slow wave, the action potentials occur, followed by contraction.

3. The basic electrical rhythm or slow waves recorded from intestinal smooth muscle:

(A) Is thought to result from ACh release from parasympathetic nerves in both longitudinal and circular muscle
(B) Is synonymous with generator potentials
(C) Varies in rate in different regions of the gastrointestinal system
(D) All of the above
(E) A and C
(F) A and B
(G) B and C

Answer: G. The basic electrical rhythm or slow waves are generator potentials, which are resulted from the regional pacemaker activity of the interstitial Cells of Cajal (ICC's). BER is an intrinsic characteristic of these cells and each region will have a different rate or rhythm.

4. Which stimulus is most important for the regulation of emptying of the intestine during the inter-digestive period?

(A) Secretin
(B) Histamine
(C) Cholecystokinin (CCK)
(D) Bombesin
(E) Motilin

Answer: E. Motilin, a hormone released from the small intestine, increases the strength of the migrating motor complex (MMC) and may be responsible for initiating it. The MMC, which occurs every 60-90 minutes during the interdigestive period, starts in the stomach and sweeps along the entire gastrointestinal (GI) tract. It is thought to be responsible for emptying the small intestine of any chyme remaining after the completion of a meal.

5. Migrating motility complexes (MMC) occur about every 90 min between meals and are thought to be stimulated by the gastrointestinal hormone, motilin. An absence of MMCs causes an increase in which of the following?

(A) Duodenal motility
(B) Gastric emptying
(C) Intestinal bacteria
(D) Mass movements
(E) Swallowing

Answer: C. Migrating motility complexes (sometimes called interdigestive myoelectric complexes) are peristaltic waves of contraction that begin in the stomach and slowly migrate in an aboral direction along the entire small intestine to the colon. By sweeping undigested food residue from the stomach, through the small intestine, and into the colon, MMCs function to maintain low bacterial counts in the upper intestine. Bacterial overgrowth syndrome can occur when the normally low bacterial colonization in the upper gastrointestinal tract increases significantly. It should be clear that an absence of MMCs would decrease duodenal motility and gastric emptying. MMCs do not have a direct effect on mass movements and swallowing.

6. A 17-year-old adolescent boy who is being treated with the macrolide antibiotic erythromycin complains of nausea, intestinal cramping, and diarrhea. The side effects are the result of the antibiotic binding to receptors in the GI tract that recognize which gastrointestinal hormone?

(A) Gastrin
(B) Secretin
(C) Cholecystokinin
(D) Motilin
(E) Enterogastrone

Answer: D. Motilin is the gastrointestinal peptide hormone associated with the initiation of migrating motor complexes during the interdigestive period. The hormone stimulates increased contractions by a direct action on smooth muscle and by activation of excitatory enteric nerves. Erythromycin belongs to the group of macrolide antibiotics and also shows an ability to excite motilin-like receptors on enteric nerves and smooth muscle. As a result, a common side effect of the antibiotic is abdominal cramping and diarrhea.

ANS

7. Parasympathetic stimulation increases gastrointestinal motility and sympathetic stimulation decreases motility. The autonomic nervous system controls gut motility by changing which of the following?

(A) Gastrin secretion
(B) Pacemaker discharge frequency
(C) Secretin secretion
(D) Slow wave frequency
(E) Spike potential frequency

Answer: E. Gastrointestinal smooth muscle undergoes rhythmical changes in membrane potential called slow waves. The slow waves are thought to be caused by variations in the sodium conductance of specialized pacemaker cells, called interstitial cells of Cajal. The discharge frequency of the pacemaker cells and hence the frequency of slow waves is fixed (i.e., does not change) in various parts of the gut. Slow-wave frequency averages about 3 per minute in the stomach, 12 per minute in the duodenum, 10 per minute in the jejunum, and 8 per minute in the ileum. When a slow wave depolarizes sufficiently, it elicits spike potentials which are true action potentials. In the small intestine, slow waves cannot initiate smooth muscle contraction in the absence of spike potentials; however, slow waves themselves can initiate the contraction of smooth muscle in the stomach. The number of spike potentials associated with a given slow wave is increased by pare-sympathetic stimulation and decreased by sympathetic stimulation. Therefore, the autonomic nervous system controls gut motility by changing the frequency of spike potentials. Neither gastrin nor secretin has significant effects on gut motility at physiological concentrations.

8. All of the following would generally depolarize intestinal smooth muscle EXCEPT:

(A) Distension
(B) Adrenergic transmitters
(C) Cholinergic agents
(D) Myenteric plexus activation
(E) Vagal discharge

Answer: B. Depolarization of a smooth-muscle cell is an excitatory stimulus. From the list provided, this would include distension. An inherent property of smooth-muscle cells is that a stretch causes depolarization and a contraction. Parasympathetics releasing ACh, and other cholinergic drugs, also depolarize and increase activity, much of this working through the myenteric plexus. On the other hand, adrenergic activity to the gut inhibits smooth muscle and decreases activity.

Esophagus

9. Swallowing is a complex process that involves signaling between the pharynx and swallowing center in the brainstem. Which of the following structures is critical for determining whether a bolus of food is small enough to be swallowed?

(A) Epiglottis
(B) Larynx
(C) Palatopharyngeal folds
(D) Soft palate
(E) Upper esophageal sphincter

Answer: C. The palatopharyngeal folds located on each side of the pharynx are pulled medially forming a sagittal slit through which the bolus of food must pass. This slit performs a selective function, allowing food that has been masticated sufficiently to pass by but impeding the passage of larger objects. The soft palate is pulled upward to close the posterior nares, which prevents food from passing into the nasal cavities. The vocal cords of the larynx are strongly approximated during swallowing and the larynx is pulled upward and anteriorly by the neck muscles. The epiglottis then swings backward over the opening of the larynx. The upper esophageal sphincter relaxes, allowing food to move from the posterior pharynx into the upper esophagus.

10. The major stimulus for primary peristalsis in the esophagus is…

(A) Presence of food in the esophagus
(B) Swallowing
(C) Regurgitation of food from the stomach
(D) Closing of the upper esophageal sphincter (UES)
(E) Opening of the lower esophageal sphincter (LES)

Answer: B. Primary esophageal peristalsis is part of the swallowing response and occurs whether or not food enters the esophagus. The intensity of the peristalsis, however, increases if food is present. If the esophagus is not emptied by primary peristalsis, the presence of food in the esophagus will initiate another peristaltic reflex, called secondary peristalsis.

11. The lower esophageal sphincter normally:

(A) Reduces its tone under the influence of gastrin
(B) Increases its tone from the parasympathetic release of vasoactive intestinal peptide
(C) Is composed of both skeletal and smooth muscle
(D) Has a yield pressure greater than intragastric pressure

Answer: D. The lower esophageal sphincter should always maintain a yield pressure greater than intragastric pressure to prevent reflux of stomach contents into the esophagus. After a meal gastrin is released and there is an increase in stomach motility and intragastric pressure. To prevent esophageal reflux, gastrin also increases the tone in the lower esophageal sphincter. VIP is an inhibitory parasympathetic transmitter released during swallowing to relax the lower sphincter. The lower sphincter is smooth muscle, the upper esophageal sphincter is skeletal muscle.

12. Acetylcholine (ACh) is required for the contraction of…

(A) The lower esophageal sphincter (LES)
(B) The upper esophageal sphincter (UES)
(C) Both the LES and UES
(D) The antrum

Answer: B. The upper esophageal sphincter (UES) is composed of striated muscle and is stimulated by cholinergic [acetylcholine (ACh)-releasing] vagal fibers. The lower esophageal sphincter (LES) is composed of smooth muscle, which normally is maintained in a contracted state by a myogenic process. ACh regulates the force of peristaltic contractions in the antrum but is not required to stimulate them.

13. A 53-year-old man complains of a mild chronic cough and heartburn. Esophageal manometric and endoscopic evaluation reveal a hypotensive lower esophageal sphincter (LES) pressure and mild gastroesophageal reflux. Which of the following is the primary genesis of LES pressure in adults?

(A) Tonic excitatory sympathetic nerve input to the smooth muscle
(B) Tonic excitatory parasympathetic nerve input to the smooth muscle
(C) Circulating gastrin
(D) Myogenic properties of LES smooth muscle
(E) Local production of nitric oxide by enteric nerves

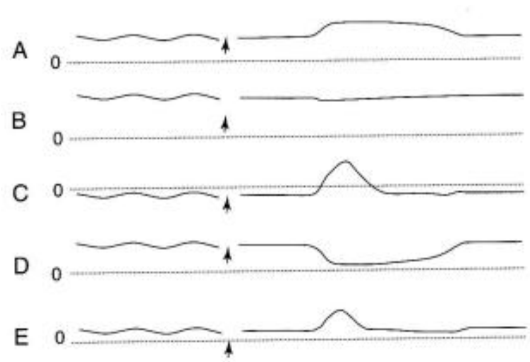
Answer: D. The lows esophageal sphincter is a high-pressure zone that exists between the esophageal body and the gastric fundus. The high pressure limits reflux of gastric contents into the esophageal body. Although excitatory vagal input contributes to the high-pressure zone, the principal determinant is intrinsic (myogenic) properties of the circular smooth muscle of the sphincter. Excess acid in the esophagus creates the pain sensation referred to as heartburn.

14. Which of the following is the putative inhibitory neurotransmitter responsible for relaxation of gastrointestinal smooth muscle?

(A) Dopamine
(B) Vasoactive intestinal peptide
(C) Somatostatin
(D) Substance P
(E) Acetylcholine

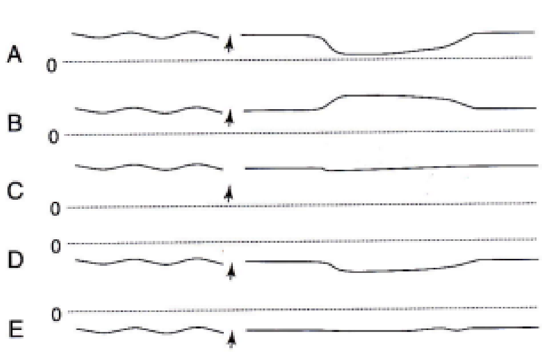
Answer: B. Important inhibitory neurotransmitters in the gastrointestinal tract include vasoactive intestinal peptide and nitric oxide. Relaxation of gastrointestinal smooth muscle occurs following activation of non-adrenergic, non-cholinergic (NANC) enteric nerve fibers. Acetylcholine, substance P, and dopamine are excitatory neurotransmitters. Somatostatin is a paracrine secretory product with multiple effects on gastrointestinal function.

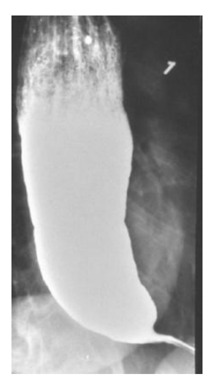
15. Which one of the following manometric recordings illustrate normal function of the esophagus at mid-thoracic level before and after swallowing (indicated by arrow)? The dashed lines represent a pressure of 0 mm Hg.



Answer: C. Trace C shows a basal subatmospheric pressure with a positive pressure wave cause by passage of the food bolus. Trace A does not correspond to any normal event in the esophagus. Trace B could represent the lower esophageal sphincter (LES) in a patient with achalasia. Trace D depicts normal operation of the LES. Trace E show a basal positive pressure trace, which does not occur where the esophagus passes through the chest cavity.

16. A 33-year-old man comes to the physician because his chest hurts when he eats, especially when he eats meat. He also belches excessively and has heartburn. His wife complains about bad breath. X-ray after barium swallow shows a dilated esophagus with narrowed LES. Which one of the pressure tracings was most likely taken at the lower esophageal sphincter of this patient before and after swallowing (indicated by arrow)? The dashed line represents a pressure of 0 mm Hg.





Answer: C. Achalasia is a condition in which the lower esophageal sphincter (LES) fails to relax during swallowing. As a result, food swallowed into the esophagus then fails to pass from the esophagus into the stomach. Trace C shows a high, positive pressure that fails to decrease after swallowing, which is indicative of achalasia. Trace A shows a normal pressure tracing at the level of the LES reflecting typical receptive relaxation in response to the food bolus. Trace E is similar to trace C, but the pressures are subatmospheric. Subatmospheric pressures occur only in the esophagus where it passes through the chest cavity.

17. A patient presents with a chronic cough. The history and physical findings rule out postnasal drip, asthma, and other pulmonary disease. Upon questioning, the patient also reports substernal burning pain that is most pronounced after ingestion of coffee, chocolate, french fries, and alcohol. Which of the following is the most likely cause of gastroesophageal reflux disease (GERD) in this patient?

(A) Delayed gastric emptying
(B) Decreased esophageal motility
(C) Hiatal hernia
(D) Decreased lower esophageal sphincter tone
(E) Decreased upper esophageal sphincter tone

Answer: D. Delayed gastric emptying, hiatal hernia, and decreased esophageal motility are all causes of GERD, but the most likely cause of GERD in this patient is a relaxed or incompetent lower esophageal sphincter (LES), which allows the gastric contents to reflux into the esophagus. The hydrochloric acid from the stomach irritates the esophageal walls, producing the substernal pain of indigestion, called heartburn. Causes of decreased LES tone include alcohol, cigarettes, coffee (caffeine), and chocolate, as well as certain drugs (nitrates and calcium channel blockers) and hormones (estrogen and progesterone). The LES is comprised of smooth muscle at the junction of the esophagus and the stomach. The upper esophageal sphincter is located between the pharynx and the esophagus and is comprised of skeletal muscle.

18. Normally, which of the following is most likely regarding reflux of gastric acid into the esophagus?

(A) It initiates primary esophageal peristalsis
(B) It inhibits gastric acid secretion
(C) It initiates secondary esophageal peristalsis
(D) It inhibits esophageal bicarbonate secretion
(E) It inhibits gastric motility

Answer: C. Reflux of gastric contents into the smooth-muscle region of the esophagus leads to the development of secondary esophageal peristalsis, characterized by enteric nerve-initiated peristalsis, beginning at the site of irritation and LES relaxation. Primary peristalsis is initiated by the medullary swallowing center and is preceded by an oral-pharyngeal phase.

19. Which of the following structures undergoes receptive relaxation when a bolus of food is swallowed?

(A) Orad stomach
(B) Palatopharyngeal folds
(C) Pharynx
(D) Thoracic esophagus
(E) Upper esophageal sphincter

Answer: A. During a swallow, the orad portion of the stomach and lower esophageal sphincter relax at about the same time. Intraluminal pressures in both regions decrease before the arrival of the swallowed bolus. This phenomenon is called receptive relaxation. Because the orad stomach relaxes with each swallow, the stomach can accept a large volume of food with only a few mm Hg rise in intragastric pressure. Receptive relaxation is mediated by afferent and efferent pathways in the vagus. Receptive relaxation and gastric distensibility are impaired following vagotomy. The palatopharyngeal folds are important for determining whether a bolus of food is small enough to be swallowed. The pharynx and thoracic esophagus undergo peristaltic contractions during swallowing, but they do not undergo receptive relaxation. The upper esophageal sphincter opens during a swallow, but this is not considered to be receptive relaxation.

Vomiting

20. Vomiting is a complex process that requires coordination of numerous components by the vomiting center located in the medulla. Which of the following occurs during the vomiting act?

	Lower Esophageal Sphincter	Upper Esophageal Sphincter	Abdominal Muscles	Diaphragm
(A)	Contract	Contract	Contract	Contract
(B)	Contract	Contract	Relax	Relax
(C)	Relax	Contract	Contract	Relax
(D)	Relax	Relax	Contract	Contract
(E)	Relax	Relax	Relax	Relax

Answer: D. The act of vomiting is preceded by antiperistalsis that may begin as far down in the gastrointestinal tract as the ileum. Distension of the upper portions of the gastrointestinal tract (especially the duodenum) becomes the exciting factor that initiates the actual act of vomiting. At the onset of vomiting, strong contractions occur in the duodenum and stomach along with partial relaxation of the lower esophageal sphincter. From then on, a specific vomiting act ensues that involves (a) a deep breath, (b) relaxation of the upper esophageal sphincter, (c) closure of the glottis, and (d) strong contractions of the abdominal muscles and diaphragm.

Stomach

21. Efferent impulses from vagal fibers to stomach may mediate:

(A) Increase strength of gastric peristalsis
(B) Relaxation of the gastric fundus during swallowing
(C) Increased secretion from parietal cells
(D) Vasoconstriction in gastric mucosa

Answer: A, B and C. Vagal activity to the stomach increases after a meal and promotes both motor and secretory activity. Another action of the vagus is that it causes relaxation of the proximal stomach as part of the swallowing reflex (called receptive relaxation). Vagal or parasympathetic activity is not involved in vasoconstriction; if anything, vasodilation would be promoted.

22. The major stimulus for receptive relaxation of the stomach is…

(A) Food in the stomach
(B) Food in the intestine
(C) Secretin
(D) Cholecystokinin (CCK)
(E) Motilin

Answer: A. When food distends the orad stomach (fundus and corpus), it produces a vagovagal reflex by which non-cholinergic, non-adrenergic fibers relax the stomach. About 2 L of food can be accommodated in the stomach when receptive relaxation is at its maximum.

23. Which of the following abolishes “receptive relaxation” of the stomach?

(A) Parasympathetic stimulation
(B) Sympathetic stimulation
(C) Vagotomy
(D) Administration of gastrin
(E) Administration of vasoactive intestinal peptide (VIP)
(F) Administration of cholecystokinin (CCK)

Answer: C. "Receptive relaxation" of the orad region of the stomach is initiated when food enters the stomach from the esophagus. This parasympathetic (vagovagal) reflex is abolished by vagotomy.

24. Twenty years ago, a 65-year-old man underwent vagotomy for his refractory peptic ulcer disease. As a result, which of the following gastrointestinal motor activities will be affected most?

(A) Secondary esophageal peristalsis
(B) Distention-induced intestinal segmentation
(C) Orad stomach accommodation
(D) Migrating motor complexes

Answer: C. Orad stomach accommodation depends exclusively on an intact vago-vagal reflex. Vagal innervation of the gastrointestinal tract extends from the esophagus to the level of the transverse colon. Preganglionic fibers from cell bodies in the medulla synapse with ganglion cells located in the enteric nervous system. Distention-induced contraction of gastrointestinal smooth muscle develops as the result of long (vago-vagal) and local (enteric nerves) reflexes. The importance of long versus local reflex pathways varies along the gut. Secondary esophageal peristalsis, intestinal segmentation, and migrating motor complexes are unaffected by vagotomy.

25. A 42-year-old man develops a gastric carcinoma affecting the proximal third of his stomach. He is scheduled for a partial gastrectomy of the affected region, which will primarily affect which of the following processes?

(A) Accommodation
(B) Peristalsis
(C) Retropulsion
(D) Segmentation
(E) Trituration

Answer: A. Increases in intragastric volume normally are not associated with large increases in intragastric pressure because of receptive relaxation or the accommodation reflex, which is vagally mediated. The reflex, which is abolished by vagotomy, is a property of the orad stomach only and counterbalances the stretch-induced myogenic contraction of the gastric smooth muscle. Peristalsis, trituration (grinding), and retropulsion (mixing) are terms referring to the contractile activity and functions of the caudad stomach. Segmental contractions are the primary contractile pattern of the small intestine during the digestive period.

26. A 63-year-old woman has an intractable duodenal ulcer failing all previous treatments. After consultation with a surgeon, a laparoscopic vagotomy is performed. Subsequently, the patient experiences nausea and vomiting after ingestion of a mixed meal. Which of the following best explains her symptoms?

(A) Increased gastric emptying of solids
(B) Decreased gastric emptying of solids
(C) Increased gastric emptying of liquids
(D) Decreased gastric emptying of liquids
(E) Increased emptying of liquids and solids

Answer: B. The vagus nerve is the primary neural mediator of gastric function. Activation of distension-mediated vago-vagal reflexes in response to the presence of food in the stomach will (1) increase gastric compliance (receptive relaxation or accommodation reflex) and promote gastric retention of food, (2) increase the strength of antral peristaltic contractions necessary for trituration of solids, and (3) increase gastric acid secretion. Sectioning of the vagus nerve fibers to the antral region of the stomach will decrease the strength of contractions thereby prolonging the emptying of solids. The emptying of liquids will be unaffected.

Gastric Emptying

27. Which of the following can occur without brain stem coordination?

(A) Chewing
(B) Swallowing
(C) Primary esophageal peristalsis
(D) Vomiting
(E) Gastric emptying

Answer: E. Gastric emptying of solids occurs when the contractile activity of the stomach reduces the size of the particles within the food sufficiently for them to pass through the pyloric sphincter. Stomach contractions are elicited by reflexes initiated by antral distention and by gastrin. Although the strength of the contractions is reduced by vagotomy, contractions can still occur. Chewing, swallowing, primary (but not secondary) peristalsis, and vomiting all are coordinated by specific regions of the brain stem.

28. Gastric emptying is regulated to ensure the chyme enters the duodenum at an appropriate rate. Which of the following factors promotes gastric emptying?

(A) Anorexia nervosa
(B) Antral peristalsis
(C) Bulimia nervosa
(D) Obesity
(E) Scleroderma
(F) Type I diabetes

Answer: B. Antral peristalsis pushes chyme toward the pylorus and thus promotes gastric emptying. Other factors that promote gastric emptying include (a) decreased compliance of the stomach, (b) relaxation of the pylorus, and (c) an absence of segmentation contractions in the small bowel. Gastric emptying is thought to be slow in eating disorders such as anorexia nervosa, bulimia nervosa, and obesity. Scleroderma is a systemic disease that affects many organ systems. The symptoms result from progressive tissue fibrosis and occlusion of the microvasculature by excessive production and deposition of types I and III collagens. Deposition of fibrous tissues in the pylorus reduces gastric emptying. Gastroparesis (paralysis of the stomach) occurs in about 20% of type I diabetics. The high blood glucose is thought to damage the vagus nerve and thereby reduce gastric emptying.

29. A 29-year-old internal medicine resident has a breakfast buffet after a long night of call. The rate of gastric emptying increases with an increase in which of the following?

(A) Intragastric volume
(B) Intraduodenal volume
(C) Fat content of duodenum
(D) Osmolality of duodenum
(E) Acidity of duodenum

Answer: A. The initial rate of emptying varies directly with the volume of the meal ingested. Increasing the volume, fat content, acidity, or osmolarity of the lumen of the small intestine inhibits gastric emptying via neural, hormonal, and paracrine feedback mechanisms.

30. The rate of gastric emptying would be relatively fast after the ingestion of a meal which has a:

(A) High osmolality
(B) High caloric content
(C) High fat content
(D) Low pH
(E) High water content

Answer: E. Isotonic, neutral pH, low caloric, liquid content has faster gastric emptying.

31. A 52-year-old man with diabetes mellitus type 1 has persistent nausea and vomiting due to gastroparesis with gastroesophageal reflux disease. Which of the following best describes the function of gastric emptying?

(A) Solids empty more rapidly than liquids.
(B) Meals containing fat empty faster than carbohydrate-rich food.
(C) Hyperosmolality of duodenal contents initiates a decrease in gastric emptying.
(D) Acidification of the antrum increases gastric emptying.
(E) Vagal stimulation decreases receptive relaxation in the upper portion of the stomach.

Answer: C. Gastroparesis is delayed emptying of food from the stomach, and is a common cause of gastroesophageal reflux disease (GERD). The rate of gastric emptying depends upon neural (enterogastric reflex) and hormonal inhibitory feedback from the proximal small bowel. Gastroparesis is common in diabetes mellitus because hyperosmolality of the duodenum initiates a decrease in gastric emptying, which is probably neural in origin and is sensed by duodenal osmoreceptors. Because solids must be liquefied prior to emptying from the stomach, the gastric emptying of liquids begins before the emptying of solids. Emptying is fastest with a carbohydrate meal, and slowest after a fatty meal Acid in the antrum inhibits gastrin secretion, which may inhibit gastric motility The vagus mediates receptive relaxation, the process in which the fundus and upper portion of the body of the stomach relax in response to movement in the pharynx and esophagus in order to accommodate food that enters the stomach.

32. Gastric emptying studies performed on a 49-year-old woman who has vomiting shortly after a meal reveal a time to one-half emptying of liquids of 18 minutes (normal < 20 minutes) and a time to one-half emptying of solids to be 150 minutes (normal < 120 minutes). Which of the following best explains the data?

(A) Increased amplitude of antral contractions
(B) Colonic obstruction
(C) Pyloric stenosis
(D) Sectioning of the vagus nerves to the stomach
(E) Inflammation of the proximal small intestine

Answer: C. The emptying of solids from the stomach is determined by the strength of antral peristaltic contractions and the resistance offered by the pyloric sphincter. Either a decrease in the amplitude of the antral contractions or an increase in sphincter resistance will delay the emptying of solids from the stomach. Liquid emptying is regulated by the proximal stomach and is primarily a function of the difference between the intragastric pressure and the intraduodenal pressure.

33. Gastric emptying is tightly regulated to ensure that chyme enters the duodenum at an appropriate rate. Which of the following events promotes gastric emptying under normal physiological conditions in a healthy person?

	Tone of Orad Stomach	Segmentation Contractions in Small Intestine	Tone of Pyloric Sphincter
(A)	Decrease	Decrease	Decrease
(B)	Decrease	Increase	Decrease
(C)	Increase	Decrease	Decrease
(D)	Increase	Decrease	Increase
(E)	Increase	Increase	Increase

Answer: C. Gastric emptying is accomplished by coordinated activities of the stomach, pylorus, and small intestine. Conditions that favor gastric emptying include (a) increased tone of the orad stomach because this helps to push chyme toward the pylorus, (b) forceful peristaltic contractions in the stomach that move chyme toward the pylorus, (c) relaxation of the pylorus which allows chyme to pass into the duodenum, and (d) absence of segmentation contractions in the intestine, which can otherwise impede the entry of chyme into the intestine.

34. The gastrointestinal hormones have physiological effects that can be elicited at normal concentrations as well as pharmacological effects that require higher than normal concentrations. What is the physiological effect of the various hormones on gastric emptying?

	Gastrin	Secretin	Cholecystokinin	GLIP	Motilin
(A)	Decrease	Decrease	Decrease	Decrease	Increase
(B)	Increase	Decrease	None	Decrease	Increase
(C)	Increase	None	None	Increase	Increase
(D)	None	None	Decrease	Increase	Increase
(E)	None	None	Decrease	None	None
(F)	None	None	Increase	None	None

Answer: E. Cholecystokinin (CCK) is the only gastrointestinal hormone that inhibits gastric emptying under normal, physiological conditions. CCK inhibits gastric emptying by relaxing the orad stomach, which increases its compliance. When the compliance of the stomach is increased, the stomach can hold a larger volume of food without excess build up of pressure in the lumen. None of the gastrointestinal hormones increase gastric emptying under physiological conditions; however, gastrin, secretin, and GLIP can inhibit gastric emptying when pharmacological doses are administered experimentally.

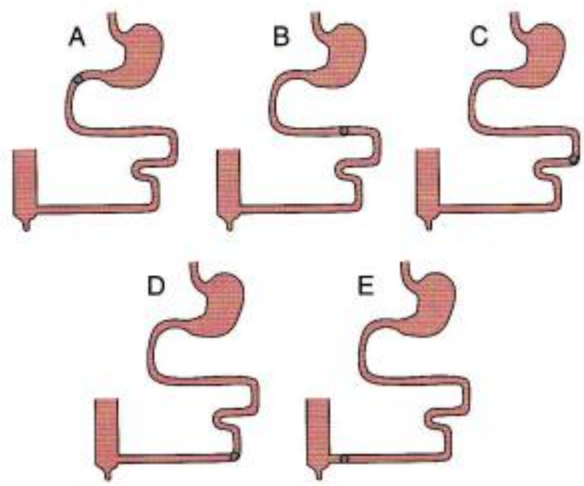
Small Intestine

35. Functions that do not need extrinsic innervation include:

(A) Initiation of swallowing
(B) Stomach emptying
(C) Defecation
(D) Mixing and propulsion in small intestine

Answer: B and D. The reflexes at the ends of the GI tract, swallowing and defecation, involve the central nervous system. But most other activity does not require central nervous system involvement. This would include stomach emptying and the mixing and propulsive movements of the small intestine.

36. A 48-year-old woman consumes a healthy meal. At which location are smooth muscle contractions most likely to have the highest frequency in the diagrams shown?



Answer: A. The frequency of slow waves is fixed in various parts of the gut. The maximum frequency of smooth muscle contractions cannot exceed the slow wave frequency. The slow wave frequency averages about 3 per minute in the stomach, 12 per minute in the duodenum, 10 per minute in the jejunum, and 8 per minute in the ileum. Therefore, the duodenum is most likely to have the highest frequency of smooth muscle contractions.

37. The motility pattern primarily responsible for the propulsion of chyme along the small intestine is…

(A) The migrating motor complex (MMC)
(B) Peristaltic waves
(C) Myogenic contractions
(D) Haustrations
(E) Segmentation

Answer: E. Although the major functions of segmentation are the mixing of chyme with digestive juices and exposing the products of digestion to the intestinal wall, segmentation is also responsible for pushing the chyme along the intestine. Propulsion occurs because the frequency of segmentation is higher in the more proximal intestine than it is in the distal intestine. Thus, it is more likely for the chyme to move toward the colon than it is to move toward the stomach. Peristaltic waves move chyme along the intestine, but these are not frequent enough to propel the food into the colon. The migrating motor complex (MMC) empties the intestine of the small amount of chyme remaining in the intestine during the interdigestive period.

38. A 24-year-old male graduate student participates in a clinical research study on intestinal motility. Peristalsis of the small intestine…

(A) Mixes the food bolus
(B) Is coordinated by the central nervous system (CNS)
(C) Involves contraction of smooth muscle behind and in front of the food bolus
(D) Involves contraction of smooth muscle behind the food bolus and relaxation of smooth muscle in front of the bolus
(E) Involves relaxation of smooth muscle simultaneously throughout the small intestine

Answer: D. Peristalsis is contractile activity that is coordinated by the enteric nervous system [not the central nervous system (CNS)] and propels the intestinal contents forward. Normally, it takes place after sufficient mixing, digestion, and absorption have occurred. ("Segmentation" is more for mixing.) To propel the food bolus forward, the smooth muscle must simultaneously contract behind it and relax in front of it.

39. The strength of peristaltic waves passing over the stomach is influenced by:

(A) Presence of 2-monoglycerides in the duodenum
(B) Distension of duodenum
(C) Osmotic pressure of chyme leaving the stomach
(D) Rate of release of adrenergic substances in the intrinsic plexus
(E) All of the above
(F) None of the above

Answer: E. There are a number of stimuli originating from stomach chyme entering the duodenum that inhibit stomach activity. This would include fat via gastric inhibitory peptide, distension initiating a nervous reflex, the high osmotic pressure of the chyme, and, although not included in this question, the presence of acid causing the release of secretin. Sympathetic adrenergic substances will also inhibit motor activity.

40. A 30-year-old man with type 1 diabetes mellitus takes metoclopramide because of severe gastroparesis. Which of the following would be expected with the use of metoclopramide?

(A) It is not associated with drug-induced parkinsonism
(B) It is a dopamine receptor agonist
(C) It decreases lower esophageal sphincter tone
(D) It increases peristalsis of the small intestine
(E) It may be used in patients with intestinal obstruction

Answer: D. Metoclopramide is used as a motility agent in patients with diabetic gastroparesis. It increases peristalsis of the small intestine and increases lower esophageal sphincter tone. It is contraindicated in patients with intestinal obstruction. Metoclopramide is a dopamine receptor antagonist and Parkinsonian-like symptoms are a known side effect of its use.

GI Reflexes

41. In a normal individual, the enterogastric reflex will be elicited by:

(A) Increased duodenal wall tension
(B) Irritation of small intestine mucosa
(C) Protein metabolites in the duodenum
(D) Acid chyme in the duodenum

Answer: A, B, C, and D. The enterogastric reflex which slows activity in the stomach will be initiated by anything that increases activity in the small intestine. The general statement is that anything that tends to overload the small intestine will act to slow the stomach. Everything listed will increase the load on the small intestine; therefore, all will contribute to the enterogastric reflex.

42. A 65-year-old man eats a healthy meal. Approximately 40 minutes later the ileocecal sphincter relaxes and chyme moves into the cecum. Gastric distention leads to relaxation of the ileocecal sphincter by way of which reflex?

(A) Enterogastric reflex
(B) Gastroileal reflex
(C) Gastrocolic reflex
(D) Intestino-intestinal reflex
(E) Rectosphincteric reflex

Answer: B. Relaxation of the ileocecal sphincter occurs with or shortly after eating. This reflex has been termed the gastroileal reflex. It is not clear whether the reflex is mediated by gastrointestinal hormones (gastrin and cholecystokinin) or extrinsic autonomic nerves to the intestine. Note that the gastroileal reflex is named with the origin of the reflex first (gastro) and the target of the reflex named second (ileal). This method of naming is characteristic of all the gastrointestinal reflexes. The enterogastric reflex involves signals from the colon and small intestine that inhibit gastric motility and gastric secretion. The gastrocolic reflex causes the colon to evacuate when the stomach is stretched. The intestino-intestinal reflex causes a bowel segment to relax when it is overstretched. The rectosphincteric reflex is also called the defecation reflex.

43. A new mother calls the pediatrician because she is concerned that her infant defecates after every meal. Which of the following is the cause of these normal bowel movements in newborns?

(A) The gastroileal reflex
(B) The gastrocolic reflex
(C) The intestino-intestinal reflex
(D) The defecation reflex
(E) Peristaltic rushes

Answer: B. Distention of the stomach by food initiates contraction of the rectum and often, a desire to defecate. This response is called the gastrocolic reflex, but it may be mediated by the action of gastrin on the colon rather than being neurally mediated. This response leads to defecation after meals in infants and children. The gastroileal reflex refers to the relaxation of the cecum and passage of chyme through the ileocecal valve when food leaves the stomach. Peristaltic rushes are very intense peristaltic waves that may occur with intestinal obstruction. The intestino-intestinal reflex refers to a complete cessation of intestinal motility that may be caused by large distensions of the intestine, injury to the intestinal wall, or various intestinal bacterial infections. The defecation reflex refers to the sudden distention of the walls of the rectum produced by mass movement of fecal material into the rectum.

44. A 54-year-old woman eats a healthy meal. Approximately 20 minutes later the woman feels the urge to defecate. Which of the following reflexes results in the urge to defecate when the stomach is stretched?

(A) Duodenocolic reflex
(B) Enterogastric reflex
(C) Gastrocolic reflex
(D) Intestino-intestinal reflex
(E) Rectosphincteric

Answer: C. The gastrocolic reflex occurs when distension of the stomach (gastro) stimulates mass movements in the colon (colic). All of the gut reflexes are named with the anatomical origin of the reflex as the prefix followed by the name of the gut segment in which the outcome of the reflex is observed, that is, the gastrocolic reflex begins in the stomach and ends in the colon. The duodenocolic reflex has a similar function to the gastrocolic reflex. When the duodenum is distended, nervous signals are transmitted to the colon, which stimulates mass movements. The enterogastric reflex occurs when signals originating in the intestines inhibit gastric motility and gastric secretion. The intestino - intestinal reflex occurs when overdistension or injury to a bowel segment signals the bowel to relax. The rectosphincteric reflex, also called the defecation reflex, is initiated when feces enters the rectum and stimulates the urge to defecate.

45. A 60-year-old woman severs her spinal cord at T6 in an automobile accident. She devises a method to distend the rectum to initiate the rectosphincteric reflex. Rectal distension causes which of the following in this woman?

	Relaxation of internal anal sphincter	Contraction of external anal sphincter	Contraction rectum
(A)	No	No	No
(B)	No	No	Yes
(C)	No	Yes	Yes
(D)	Yes	No	Yes
(E)	Yes	Yes	No
(F)	Yes	Yes	Yes

Answer: D. When feces enters the rectum, distention of the rectal wall initiates signals that spread through the myenteric plexus to initiate peristaltic waves in the descending colon, sigmoid colon, and rectum all of which force feces toward the anus. At the same time the internal anal sphincter relaxes allowing the feces to pass. In people with transected spinal cords, the defecation reflexes can cause automatic emptying of the bowel because the external anal sphincter is normally controlled by the conscious brain through signals transmitted in the spinal cord.

46. A healthy, 21-year-old woman eats a big meal and then takes a 3-hour ride on a bus that does not have a bathroom. Twenty minutes after eating, the woman feels a strong urge to defecate, but manages to hold it. Which of the following have occurred in this woman?

	Relaxation of Internal Anal Sphincter	Contraction of External Anal Sphincter	Contraction of Rectum
(A)	No	No	No
(B)	No	Yes	Yes
(C)	Yes	No	Yes
(D)	Yes	No	No
(E)	Yes	Yes	Yes

Answer: E. The defecation reflex (also called the rectosphincteric reflex) occurs when a mass movement forces feces into the rectum. When the rectum is stretched, the internal anal sphincter relaxes and the rectum contracts pushing the feces toward the anus. The external anal sphincter is controlled voluntarily and can be contracted when defecation is not possible. Therefore, when a person feels the urge to defecate, the internal anal sphincter is relaxed, the rectum is contracting, and the external anal sphincter is either contracted or relaxed depending on the circumstances.

47. Which of the following changes occurs during defecation?

(A) Internal anal sphincter is relaxed
(B) External anal sphincter is contracted
(C) Rectal smooth muscle is relaxed
(D) Intra-abdominal pressure is lower than when at rest
(E) Segmentation contractions predominate

Answer: A. Both the internal and external anal sphincters must be relaxed to allow feces to be expelled from the body. Rectal smooth muscle contracts and intra-abdominal pressure is elevated by expiring against a closed glottis (Valsalva maneuver). Segmentation contractions are prominent in the small intestine during digestion and absorption.

48. Mass movements are often stimulated after a meal by distention of the stomach (gastrocolic reflex) and distention of the duodenum (dundenocolic reflex). Mass movements often lead to which of the following?

(A) Bowel movements
(B) Gastric movements
(C) Haustrations
(D) Esophageal contractions
(E) Pharyngeal peristalsis

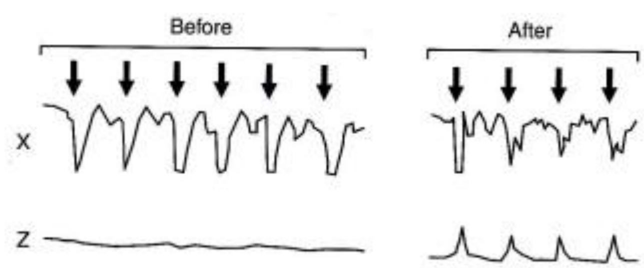
Answer: A. Mass movements force feces into the rectum. When the walls of the rectum are stretched by the feces, the defecation reflex is initiated and a bowel movement follows when this is convenient. Mass movements do not affect gastric motility. Haustrations are bulges in the large intestine caused by contraction of adjacent circular and longitudinal smooth muscle. It should be clear that mass movements in the colon do not affect esophageal contractions or pharyngeal peristalsis.

49. Mass movements constitute an important intestinal event that lead to bowel movements. Mass movements cause which of the following?

(A) Contraction of internal anal sphincter
(B) Duodenal peristalsis
(C) Gastric retropulsion
(D) Hunger sensations
(E) Rectal distension

Answer: E. The rectum is empty of feces most of the time. When a mass movement forces feces into the rectum, the desire to defecate is initiated immediately. Reflex contraction of the rectum and relaxation of the internal anal sphincter follows. If a person is in a place where defecation is possible (like a bathroom), the external anal sphincter is consciously relaxed and the feces is expelled. It should be clear that mass movements do not cause duodenal peristalsis, gastric retropulsion, or hunger sensations.

Question 50 and 51
The following diagram shows manometric recordings from a patient before and after pelvic floor training. A balloon placed in the rectum was blown up (arrows) and deflated repeatedly. Tracing Z is a manometric recording obtained from the external anal sphincter before and after pelvic floor training. (Next 2 questions)



50. Which of the following best describes the origin of tracing X shown in the upper panel?

(A) Distal rectum
(B) Ileocecal valve
(C) Internal anal sphincter
(D) Lower esophageal sphincter
(E) Proximal rectum

Answer: C. The internal anal sphincter relaxes when the rectum is stretched, as indicated by repeated decreases in pressure following inflation of the rectal balloon. Pressures in the distal and proximal rectum are expected to increase following inflation of the rectal balloon. Inflation of a rectal balloon should not affect pressures at the lower esophageal sphincter or ileocecal valve.

51. Which of the following best describes the condition for which the patient received pelvic floor training?

(A) Anal fissure (i.e., tear or superficial laceration)
(B) Chronic diarrhea
(C) Fecal incontinence (i.e., no control over defecation)
(D) Hemorrhoids
(E) Hirschsprung disease

Answer: C. Prior to pelvic floor training, the pressure at the external anal sphincter was unchanged following inflation of the rectal balloon. This failure of the external anal sphincter to contract is expected to result in defecation. After pelvic floor training, the external anal sphincter contracts when the rectal balloon is inflated, which prevents inappropriate defecation.

Hirschsprung disease

52. A newborn boy does not pass meconium in the first 24 hr. His abdomen is distended and he begins vomiting. Various tests lead to a diagnosis of Hirschsprung disease. An obstruction is most likely found in which portion of the gut?

(A) Ascending colon
(B) Ileocecal sphincter
(C) Lower esophageal sphincter
(D) Pylorus
(E) Sigmoid colon

Answer: E. Hirschsprung disease is characterized by a congenital absence of ganglion cells in the distal colon resulting in a functional obstruction. Most cases of Hirschsprung disease are diagnosed in the newborn period. Hirschsprung disease should be considered in any newborn who fails to pass meconium within 24 to 48 hr after birth. Although contrast enema is useful in establishing the diagnosis, rectal biopsy remains the criterion standard. Aganglionosis begins with the anus, which is nearly always involved, and continues proximally for a variable distance. Both the myenteric (Auerbach) and submucosal (Meissner) plexus are absent, resulting in reduced bowel peristalsis and function. The precise mechanism underlying the development of Hirschsprung disease is poorly understood.

53. A newborn boy has a distended abdomen, fails to pass meconium within the first 48 hr of life, and vomits repeatedly. Analysis of a rectal biopsy provides a definitive diagnosis of Hirschsprung disease. The absence of which type of cell is diagnostic for Hirschsprung disease?

(A) Lymphatic endothelial cells
(B) Capillary endothelial cells
(C) Parasympathetic ganglion cells
(D) Red blood cells
(E) Smooth muscle cells

Answer: C. Hirschsprung disease results from the absence of parasympathetic ganglion cells in the myenteric and submucosal plexus of the rectum and/or colon. Congenital aganglionosis begins with the anus, which is always involved, and continues proximally for a variable distance. Both the myenteric (Auerbach) and submucosal (Meissner) plexus are absent, resulting in reduced bowel peristalsis and function. The precise mechanism underlying the development of Hirschsprung disease is unknown. It should be clear that an absence of lymphatic endothelial cells, capillary endothelial cells, or red blood cells would not affect colonic motility. Smooth muscle cells are present in Hirschsprung disease.

Gastrointestinal Secretion

54. Serves as lubricant in gastrointestinal tract.

(A) HCI (B) HCO_3^- (C) Mucin (D) NaCl

Answer: C. Mucin is produced throughout the GI tract, and one function is as a lubricant.

55. Cystic fibrosis (CF) is an inherited disorder of the exocrine glands, affecting children and young people. Mucus in the exocrine glands becomes thick and sticky and eventually blocks the ducts of these glands (especially in the pancreas, lungs, and liver), forming cysts. A primary disruption in the transfer of which ion across cell membranes occurs in CF leading to decreased secretion of fluid?

(A) Calcium
(B) Chloride
(C) Phosphate
(D) Potassium
(E) Sodium

Answer: B. Movement of chloride ions out of cells leads to secretion of fluid by cells. Cystic fibrosis (CF) is caused by abnormal chloride ion transport on the apical surface of epithelial cells in exocrine gland tissues. The CF transmembrane regulator (CFTR) protein functions both as a cyclic AMP-regulated Cl^- channel and, as its name implies, a regulator of other on channels. The fully processed form of CFTR is found in the plasma membrane of normal epithelia. Absence of CFTR at appropriate cellular sites is often part of the patho-physiology of CF. However, other mutations in the CF gene produce CFTR proteins that are fully processed but are nonfunctional or only partially functional at the appropriate cellular sites.

Saliva

56. Which of the following secretions is most dependent on neural stimulation?

(A) Saliva
(B) Hydrochloric acid
(C) Pepsin
(D) Pancreatic juice
(E) Bile

Answer: A. Salivary flow is entirely dependent on the autonomic nervous system (ANS). Parasympathetic stimulation produces a large volume of watery fluid, while sympathetic stimulation causes the secretion of proteins (mucus and some enzymes). Secretion of hydrochloric acid (HCI), pepsin, pancreatic juice, and bile is influenced by vagal stimulation but can occur without it.

57. Which of the following is characteristic of saliva?

(A) Hypotonicity relative to plasma
(B) A lower HCO_3^- concentration than plasma
(C) The presence of proteases
(D) Secretion rate that is increased by vagotomy
(E) Modification by the salivary ductal cells involves reabsorption of K^+ and HCO_3^-

Answer: A. Saliva is characterized by hypotonicity and a high HCO_3^-, concentration (relative to plasma) and by the presence of α-amylase and lingual lipase (not proteases). The high HCO_3^- concentration is achieved by secretion of HCO_3^- into saliva by the ductal cells (not reabsorption of HCO_3^-). Because control of saliva production is parasympathetic, it is abolished by vagotomy.

58. Compared to fluid in a salivary gland acinus, the fluid at the duct opening in the mouth will have a:

(A) Lower concentration of potassium ions
(B) Lower osmolarity
(C) Higher concentration of sodium ions
(D) Higher concentration of chloride ions
(E) Lower pH

Answer: B. The ducts of salivary glands reabsorb Na^+ and Cl^-, and secrete K^+. H_2O is impermeable there and thus, at rest (low flow rate), saliva is hypotonic. Since HCO_3^- is not reabsorbed, the saliva is alkalinic.

59. Compared to plasma, saliva has the highest relative concentration of which of the following ions under basal conditions?

(A) Bicarbonate (B) Chloride (C) Potassium (D) Sodium

Answer: C. Under basal conditions, saliva contains high concentrations of potassium and bicarbonate ions and low concentrations of sodium and chloride ions. The primary secretion of saliva by acini has an ionic composition similar to that of plasma. As the saliva flows through the ducts, sodium ions are actively reabsorbed and potassium ions are actively secreted in exchange for sodium. Because sodium is absorbed in excess, chloride ions follow the electrical gradient causing chloride levels in saliva to decrease greatly. Bicarbonate ions are secreted by an active transport process causing an elevation of bicarbonate concentration in saliva. The net result is that, under basal conditions, sodium and chloride concentrations in saliva are about 10% to 15% of that of plasma, bicarbonate concentration is about three-fold greater than that of plasma, and potassium concentration is about seven times greater than that of plasma.

60. Which of the following ions has the highest concentration in saliva under basal conditions?

(A) Bicarbonate (B) Chloride (C) Potassium (D) Sodium

Answer: A. Although the potassium concentration in saliva is about seven times greater than that of plasma, and the bicarbonate concentration in saliva is only about three times greater than that of plasma, the actual concentration of bicarbonate in saliva is 50 to 70 mEq/L, whereas the concentration of potassium is about 30 mEq/L, under basal conditions.

61. After a recent viral illness, a 20-year-old woman develops bilateral facial swelling consistent with parotitis. Which of the following best describes the salivary glands?

(A) The parotid gland is the most mucinous of the salivary glands.
(B) Starch digestion begins in the mouth via α-amylase.
(C) α-Amylase works most efficiently in the stomach.
(D) Approximately 4 L of saliva is secreted per day.
(E) Cranial nerve VIII passes through the parotid gland.

Answer: B. Starch digestion begins in the mouth via α-amylase, which is inactivated by the acidity of the stomach. The parotid gland is the most serous of the salivary glands and cranial nerve VII, not VIII, runs through the parotid (though it does not innervate the gland). Approximately 1.5 L of saliva is secreted per day.

Stomach

62. According to the two-component theory of gastric secretion, the H^+ concentration of gastric juice increases with the rate of secretion because:

(A) The oxyntic component is hypertonic
(B) The volume of the oxyntic component increases
(C) Oxyntic cells secrete a higher $[H^+]$ than non-oxyntic cells
(D) The volume of the non-oxyntic component decreases
(E) Na^+ absorption is stimulated

Answer: B. At rest, the gastric juice comprises a non-oxyntic component derived from paracellular diffusion of interstitial fluid and secretions from mucus cells. This produces a basal alkaline fluid of constant composition and low volume. As oxyntic cells are stimulated, the non-oxyntic fluid is gradually diluted with HCl/KCl until the final $[Na^+]$ is only about 5 mM.

63. Free H^+ in the gastric juice of a mammal is:

(A) Associated with increased alkalinity of the plasma leaving the stomach
(B) Well-correlated with gastric peptic activity under physiologic conditions
(C) Controlled by both hormonal and autonomic nervous mechanisms
(D) Produced by same stomach epithelial cells secreting pepsinogen

Answer: A, B, and C. The secretion of one H^+ into the stomach is associated with the secretion of one HCO_3^- into the venous system. This causes stomach venous blood to become more alkaline. Also, after a meal the demand for CO_2 by the parietal cells as a source of H^+ is so high that it extracts CO_2 from capillary blood. Thus, venous blood draining the stomach after a meal is low in CO_2 and high in HCO_3^-, which makes for a very alkaline state. There is generally a good correlation between secretion of the parietal cells (secreting H^+) and chief cells (secreting pepsin) in the gastric glands. Hormonal control is via gastrin (stimulates) and secretin (inhibits). Nervous control is via the release of acetylcholine (stimulates) from parasympathetic nerve endings.

64. The major stimulus for gastric acid (HCl) secretion during the cephalic phase is…

(A) Histamine
(B) Gastrin
(C) Secretin
(D) Somatostatin
(E) Acetylcholine (ACh)

Answer: E. During the cephalic phase of gastric acid (HCl) secretion, the sight, smell, or thought of food activates cholinergic [acetylcholine (ACh)-releasing] vagal fibers, which stimulate the release of HCl from antral parietal cells.

65. Vagal stimulation plays an essential role during the cephalic and gastric phases of gastric secretion. Vagal stimulation tends to cause which of the following changes in the release of gastric releasing peptide (GRP) and somatostatin?

	GRP	Somatostatin
(A)	↑	↑
(B)	↓	↓
(C)	↑	↓
(D)	↓	↑
(E)	↔	↓

Answer: C. The cephalic phase of gastric secretion is mediated entirely by the vagus nerve; vagotomy abolishes the response. The vagus also mediates a significant portion of the gastric phase of gastric secretion. Vagal stimulation increases acid secretion by a direct action of parietal cells as well as by stimulating gastrin secretion. Vagal stimulation increases gastrin secretion by (a) directly stimulating G cells, which secrete gastrin, and (b) inhibiting somatostatin cells from secreting stomatostatin, which would otherwise inhibit G cells from secreting gastrin. Gastrin releasing peptide (GRP) is the neurotransmitter released from interneurons that stimulate G cells to secrete gastrin.

66. The cephalic phase of gastric secretion accounts for about 30% of the acid response to a meal. Which of the following can totally eliminate the cephalic phase of gastric secretion?

(A) Antacids (e.g., Rolaids)
(B) Anti-gastrin antibody
(C) Atropine
(D) Histamine H_2 blocker
(E) Vagotomy
(F) Sympathectomy

Answer: E. The cephalic phase of gastric secretion occurs before food enters the stomach. Seeing, smelling, chewing, and anticipating food is perceived by the brain, which, in essence, tells the stomach to prepare for a meal. Stimuli for the cephalic phase thus include mechanoreceptors in the mouth, chemoreceptors (smell and taste), thought of food, and hypoglycemia. Because the cephalic phase of gastric secretion is mediated entirely by way of the vagus nerve, vagotomy can abolish the response. Antacids neutralize gastric acid, but they do not inhibit gastric secretion. An antigastrin antibody would attenuate (hut not abolish) the cephalic phase because this would have no effect on histamine and acetylcholine stimulation of acid secretion. Atropine would attenuate the cephalic phase by blocking acetylcholine receptors on parietal cells; however, atropine does not abolish acetylcholine stimulation of gastrin secretion. A histamine H_2 blocker would attenuate the cephalic phase of gastric secretion, but would not abolish it.

67. Gastric acid secretion increases when food enters the stomach because…

(A) Protein digestion products indirectly stimulate the parietal cells to release hydrochloric acid (HCI)
(B) Food raises the pH of the stomach, allowing more acid to be released
(C) Both
(D) Neither

Answer: C. Gastric acid secretion is increased indirectly by protein digestion products and inhibited when the pH of the stomach is reduced. The buffering action of food promotes gastric acid secretion by keeping the pH from falling too low.

68. Which stimulus is most important for the regulation of gastric acid (HCl) secretion?

(A) Secretin
(B) Histamine
(C) Cholecystokinin (CCK)
(D) Bombesin (GRP)
(E) Motilin

Answer: B. Gastric acid (HCl) secretion is stimulated by acetylcholine (ACh), histamine, and gastrin. It is inhibited by somatostatin. GRP stimulates G cells to release gastrin.

69. Which stimulus is most important for the regulation of gastrin secretion?

(A) Secretin
(B) Histamine
(C) Cholecystokinin (CCK)
(D) Bombesin (GRP)
(E) Motilin

Answer: D. Gastrin secretion is stimulated by bombesin [also called gastrin-releasing peptide [GRP)] and inhibited by somatostatin.

70. Increases stomach acid secretion.

(A) Gastrin
(B) Secretin
(C) Cholecystokinin
(D) Gastric inhibitory peptide

Answer: A. A major target tissue for gastrin is the stomach, where it stimulates both secretion and motility. Secretin inhibits stomach secretion and motility.

71. A 42-year-old salesman presents with the chief complaint of intermittent midepigastric pain that is relieved by antacids or eating. Gastric analysis reveals that basal and maximal acid outputs exceed normal values. The gastric acid hypersecretion can be explained by an increase in the plasma concentration of which of the following?

(A) Somatostatin
(B) Vasoactive intestinal peptide
(C) Gastrin
(D) Secretin
(E) Cholecystokinin

Answer: C. Increases in basal and maximal acid output are suggestive of inflammation or removal of the proximal small intestine. Intestinal receptors monitor the composition of chyme and elicit feedback mechanisms that regulate gastric acid secretion and gastric emptying. Absence of feedback leads to an increased presence of excitatory mediators of gastric function. Gastrin is the primary stimulus of meal-induced acid secretion by the parietal cells. Somatostatin, vasoactive intestinal peptide, secretin, and cholecystokinin inhibit gastric acid secretion.

72. A 42-year-old man is referred to a gastroenterologist for evaluation of refractory peptic ulcer disease. Subsequent endoscopic and laboratory data are suggestive of Zollinger-Ellison syndrome. The increased basal acid output and maximal acid output of the patient is best explained by an increase in the plasma concentration of which of the following?

(A) Gastrin-releasing peptide
(B) Secretin
(C) Somatostatin
(D) Gastrin
(E) Histamine

Answer: D. Zollinger-Ellison (ZE) has a triad of peptic ulcer disease, gastric acid hypersecretion, and an elevated gastrin level. In ZE, a pancreatic acinar cell adenoma (gastrinoma) is the site for the synthesis and secretion of large amounts of gastrin. Unlike gastrin released from the antrum in response to normal physiological stimuli, the pancreatic release of gastrin from the pancreas is not under physiological control, that is, intestinal feedback and gastric pH. ZE can be part of a multiple endocrine tumor (MEN I). Proton pump inhibitors (omeprazole, lansoprazole) are the treatment of choice for peptic ulcer disease in ZE, and have decreased the need for total gastrectomy.

73. An 82-year-old woman with upper abdominal pain and blood in the stool has been taking NSAIDS for arthritis. Endoscopy revealed patchy gastritis throughout the stomach. Biopsies were negative for *Halicobacter pylori*. Pentagastrin administered intravenously would lead to a less than expected (i.e., less than normal) increase in which of the following?

(A) Duodenal mucosal growth
(B) Gastric acid secretion
(C) Gastrin secretion
(D) Pancreatic enzyme secretion
(E) Pancreatic growth

Answer: B. The use of nonsteroidal anti-inflammatory drugs (NSAIDs) may result in NSAID-associated gastritis or peptic ulceration. Chronic gastritis, by definition, is a histopathological entity characterized by chronic inflammation of the stomach mucosa. When inflammation affects the gastric corpus, parietal cells are inhibited, leading to reduced acid secretion. Although diagnosis of chronic gastritis can only be ascertained histologically, the administration of pentagastrin should produce a less than expected increase in gastric acid secretion. Pentagastrin is a synthetic gastrin composed of the terminal four amino acids of natural gastrin plus the amino acid alanine. It has all the same physiologic properties of natural gastrin. Although gastrin and pentagastrin can both stimulate growth of the duodenal mucosa, it should be clear that intravenous pentagastrin would not cause substantial growth in the context of a clinical test. In any case, chronic administration of pentagastrin would not lead to a less than expected growth of the duodenal mucosa. Pentagastrin is not expected to increase gastrin secretion, pancreatic enzyme secretion, or pancreatic growth.

74. Which of the following is the site of secretion of gastrin?

(A) Gastric antrum
(B) Gastric fundus
(C) Duodenum
(D) Ileum
(E) Colon

Answer: A. Gastrin is secreted by the G cells of the gastric antrum. HCl and intrinsic factor are secreted by the fundus.

75. Where does the secretion of gastrin occur?

(A) Fundus of the stomach
(B) Antrum of the stomach
(C) Duodenum of the intestine
(D) Ileum of the intestine
(E) Colon

Answer: B. Gastrin is secreted by G cells contained in the pyloric glands of the distal stomach (antrum and pylorus).

76. Regulates gastrin output in stomach.

(A) HCl
(B) HCO_3^-
(C) Mucin
(D) NaCl

Answer: A. Stomach acid acts a negative feedback loop on the release of gastrin. Lack of stomach acid production produces high circulating gastrin.

77. Secretion of which of the following substances is inhibited by low pH?

(A) Secretin
(B) Gastrin
(C) Cholecystokinin (CCK)
(D) Vasoactive intestinal peptide (VIP)
(E) Gastric inhibitory peptide (GIP)

Answer: B. Gastrin's principal physiologic action is to increase H^+ secretion. H^+ secretion decreases the pH of the stomach contents. The decreased pH, in turn, inhibits further secretion of gastrin-a classic example of negative feedback.

78. Low gastric pH:

(A) Inhibits release of gastrin
(B) Inhibits release of histamine
(C) Is necessary for pepsin activity
(D) Reduces irritation caused by aspirin

Answer: A, B, and C. A low gastric pH acts as a negative feedback loop to inhibit the release of gastrin. If the stomach becomes alkaline, the acid suppression of gastrin release is eliminated and the result is high circulating gastrin. The local release of histamine, which stimulates the parietal cells, is also suppressed by stomach acid. The protease pepsin has an acid pH optimum and thus can only function in the stomach. When chyme enters the duodenum, pepsin is no longer active.

79. A 35-year-old male smoker presents with burning epigastric pain that is most pronounced on an empty stomach. A paroxysmal rise in serum gastrin in response to intravenous secretin further supports a diagnosis of Zollinger-Ellison syndrome. Normally, basal acid output is increased by which of the following?

(A) Acidification of the antrum
(B) Administration of an H_2-receptor antagonist
(C) Vagotomy
(D) Alkalinization of the antrum
(E) Acidification of the duodenum

Answer: D. Alkalinization of the antrum releases the gastrin-containing cells from the inhibitory influences of somatostatin and increases acid secretion. Acidification of the antrum promotes the release of somatostatin from cells in the gastrointestinal mucosa, which inhibits gastrin release and gastric acid secretion. Acidification of the duodenum elicits inhibitory neural and hormonal reflexes that also inhibit acid output. Administration of a histamine antagonist, proton pump inhibitor, and vagotomy all reduce acid secretion.

80. Eating a meal leads to a large increase in gastric acid secretion that peaks within about 5 minutes and returns to normal about 4 hours after a meal is taken. How long after a meal does the pH of the gastric contents reach its lowest level (in hours)?

(A) 1.0 (B) 1.5 (C) 2.0 (D) 2.5 (E) 3.0 (F) 4.0

Answer: F. It is a common misconception that the pH of the gastric contents is lowest (most acidic) following a meal when acid secretion is highest. Before a meal, when the stomach is empty, the pH of the gastric contents is the lowest and acid secretion is suppressed. Acid secretion is low because (a) the acid stimulates somatostatin release (which has a direct action to decrease secretion of both gastrin and acid), and (b) acid has a direct effect to suppress parietal cell secretions. When a meal is taken, the buffering effects of the food cause the gastric pH to increase, which allows the various secretagogues to stimulate acid secretion.

81. The control of gastric acid secretion in response to a meal involves several events that take place over a 4- or 5-hr period following the meal. These events include (1) a decrease in the pH of the gastric contents, (2) an increase in the rate of acid secretion, (3) a decrease in the rate of acid secretion, and (4) an increase in the pH of the gastric contents. Which of the following best describes the correct temporal order of events over a 4- or 5-hr period following a meal?

(A) 4, 3, 2, 1 (C) 3, 4, 1, 2 (E) 4, 2, 1, 3 (G) 2, 3, 1, 4
(B) 3, 1, 4, 2 (D) 2, 1, 4, 3 (F) 1, 2, 3, 4 (H) 1, 3, 2, 4

Answer: E. Following a meal, the pH or the gastric contents increases because the food buffers the acid in the stomach. This increase in pH suppresses the release of somatostatin from delta cells in the stomach (hydrogen ions stimulate the release of somatostatin). Because somatostatin inhibits secretion of both gastrin and gastric acid, the fall in somatostatin levels leads to an increase in acid secretion. The increase in acid secretion causes the pH of the gastric contents to decrease. As the pH of the gastric contents decreases, the rate of acid secretion also decreases.

82. Gastric parietal cells secrete…

(A) Gastrin
(B) Motilin
(C) Cholecystokinin (CCK)
(D) Intrinsic factor
(E) Secretin

Answer: D. Parietal cells, located in the oxyntic glands of the orad stomach (fundus and corpus), secrete hydrochloric acid (HCl) and intrinsic factor. Gastrin is secreted by G cells located in the pyloric glands of the distal stomach (antrum). Secretin and cholecystokinin (CCK) are secreted by endocrine cells in the proximal intestine. Motilin is secreted from endocrine cells within the epithelium of the small intestine.

83. When parietal cells are stimulated, they secrete

(A) HCl and intrinsic factor
(B) HCl and pepsinogen
(C) HCl and HCO_3^-
(D) HCO_3^- and intrinsic factor
(E) Mucus and pepsinogen

Answer: A. The gastric parietal cells secrete HCl and intrinsic factor. The chief cells secrete pepsinogen.

84. A 71-year-old man with upper abdominal pain and blood in the stool takes NSAIDS for the pain and washes it down with whiskey. Pentagastrin administration produced lower than predicted levels of gastric acid secretion. Secretion of which of the following substances is most likely to be diminished in this patient with gastritis?

(A) Intrinsic factor
(B) Ptyalin
(C) Rennin
(D) Saliva
(E) Trypsin

Answer: A. Intrinsic factor is a glycoprotein secreted from parietal cells (i.e., acid secreting cells in the stomach) that is necessary for absorption of vitamin B12. The patient has a diminished capacity to secrete acid because of chronic gastritis. Because acid and intrinsic factor are both secreted by parietal cells, a diminished capacity to secrete acid is usually associated with diminished capacity to secrete intrinsic factor. Ptyalin, also known as salivary amylase, is an enzyme that begins carbohydrate digestion in the mouth. The secretion of ptyalin is not affected by gastritis. Rennin, known also as chymosin, is a proteolytic enzyme synthesized by chief cells in the stomach. Its role in digestion is to curdle or coagulate milk in the stomach, a process of considerable importance in very young animals. It should be clear that saliva secretion is not affected by gastritis. Trypsin is a proteolytic enzyme secreted by the pancreas.

85. Where does the secretion of intrinsic factor occur?

(A) Fundus of the stomach
(B) Antrum of the stomach
(C) Duodenum of the intestine
(D) Ileum of the intestine
(E) Colon

Answer: A. Intrinsic factor, necessary for the absorption of vitamin B_{12}, is secreted by parietal cells located in the proximal stomach (fundus).

86. Which of the following is the site of secretion of intrinsic factor?

(A) Gastric antrum
(B) Gastric fundus
(C) Duodenum
(D) Ileum
(E) Colon

Answer: B. Intrinsic factor is secreted by the parietal cells of the gastric fundus (as is HCl). It is absorbed, with vitamin B_{12}, in the ileum.

87. Biopsies are taken from the antral and duodenal mucosa of a 65-year-old woman. Which of the following hormones can be found in tissue homogenates from both locations?

(A) Cholecystokinin (CCK)
(B) Glucose-dependent insulinotropic peptide (GLIP)
(C) Gastrin
(D) Motilin
(E) Secretin

Answer: C. The gastrointestinal hormones are secreted from endocrine cells located in the mucosa. The endocrine cells are not clumped together but are dispersed among the epithelial cells, making it virtually impossible to remove surgically the source of any one gastrointestinal hormone. Gastrin is the only listed hormone found in the antrum, but it is also found in the duodenal and to a lesser extent the jejunal mucosa. CCK and secretin secreting endocrine cells are found in the duodenum, jejunum, and ileum. Moulin and GLIP secreting cells are found in the duodenum and jejunum.

88. The proenzyme pepsinogen is secreted mainly from which of the following structures?

(A) Acinar cells of the pancreas
(B) Ductal cells of the pancreas
(C) Epithelial cells of the duodenum
(D) Gastric glands of the stomach

Answer: D. Pepsinogen is the precursor of the enzyme pepsin. Pepsinogen is secreted from the peptic or chief cells of the gastric gland (also called the oxyntic gland). To be converted from the precursor form to the active form (pepsin), pepsinogen must come in contact with hydrochloric acid or pepsin itself. Pepsin is a proteolytic enzyme that digests collagen and other types of connective tissue in meats.

Gastric Mucosa

89. The gastric mucosal barrier has a physiological and an anatomical basis to prevent back-leak of hydrogen ions into the mucosa. Some factors are known to strengthen the integrity of the gastric mucosal barrier, whereas other factors can weaken the barrier. Which of the following factors strengthen or weaken the barrier?

	Bile salts	Mucous	Aspirin	NSAIDS	Gastrin	Ethanol
(A)	Strengthen	Strengthen	Weaken	Weaken	Strengthen	Strengthen
(B)	Strengthen	Strengthen	Weaken	Weaken	Weaken	Strengthen
(C)	Weaken	Strengthen	Strengthen	Weaken	Strengthen	Weaken
(D)	Weaken	Strengthen	Weaken	Weaken	Strengthen	Weaken
(E)	Weaken	Weaken	Weaken	Strengthen	Strengthen	Weaken

Answer: D. Damage to the gastric mucosal barrier allows hydrogen ions to back-leak into the mucosa in exchange for sodium ions. A low pH in the mucosa causes mast cells to leak histamine, which damages the vasculature causing ischemia. The ischemic mucosa allows a greater leakage of hydrogen ions—more cell injury and death—resulting in a vicious cycle. Factors that normally strengthen the gastric mucosal barrier include mucus (which impedes the influx of hydrogen ions), gastrin (which stimulates mucosal growth), certain prostaglandins (which can stimulate mucus secretion), and various growth factors that can stimulate growth of blood vessels, gastric mucosa, and other tissues. Factors that weaken the gastric mucosal barrier include *Helicobacter pylori* (a bacterium that produces toxic levels of ammonium) as well as aspirin, NSAIDS, ethanol, and bile salts.

90. A 62-year-old woman who takes an NSAID for her severe bilateral osteoarthritis of the knees is prescribed misoprostol. Which of the following would be expected with the use of misoprostol (prostaglandin E_1 analog)?

(A) Diarrhea is not associated with its use
(B) It may be safely used in pregnancy
(C) It is used in the prevention of NSAID-induced gastric ulcers
(D) It is a histamine receptor antagonist
(E) It inhibits mucous bicarbonate secretion

Answer: C. Misoprostol is used in the prevention of NSAID-induced gastric ulcers. It helps maintain the gastric mucosal barrier, which can become compromised with prostaglandin inhibitors such as NSAIDs. Misoprostol enhances mucous bicarbonate secretion in the stomach. Diarrhea is the most common side effect and its use is contraindicated in pregnancy

91. A 45-year-old man presents with abdominal pain and hematemesis. An abdominal exam was relatively benign, and abdominal x-rays were suggestive of a perforated viscus. Endoscopy revealed a chronically perforated gastric ulcer, through which the liver was visible. Which of the following is a forerunner to gastric ulcer formation?

(A) Back-leak of hydrogen ions
(B) Mucus secretion
(C) Proton pump inhibition
(D) Tight junctions between cells
(E) Vagotomy

Answer: A. Hydrogen ions leak into the mucosa when it is damaged. As the hydrogen ions accumulate in the mucosa, the intracellular buffers become saturated, and the pH of the cells decreases resulting in injury and cell death. The hydrogen ions also damage mast cells causing them to secrete excess amounts of histamine. The histamine exacerbates the condition by damaging blood capillaries within the mucosa. The result is focal ischemia, hypoxia, and vascular stasis. The mucosal lesion is a forerunner of gastric ulcer. Mucus secretion helps to strengthen the gastric mucosal barrier because mucus impedes the leakage of hydrogen ions into the mucosa. Various proton pump inhibitors are used as a treatment modality for gastric ulcer because these can decrease the secretion of hydrogen ions (protons). The tight junctions between cells within the mucosa help to prevent the back-leak of hydrogen ions. Vagotomy was once used to treat gastric ulcer disease because severing or crushing the vagus nerve decreases gastric acid secretion.

Gastric Ulcer

92. Your patient is suffering from a gastric ulcer. Which one of the following would be the most likely to exacerbate this affliction?

(A) Overuse of histamine-antagonists
(B) Stimulation of the Chief cells
(C) Inhibition of the parietal cells
(D) Stimulation of the ECL cells
(E) Stimulation of D cells

Answer: B and D. Main causative problem of gastric ulcer is high $[H^+]$. Pepsin promotes tissue injury (autodigestion).

Helicobader pylori

93. Damage to the gastric mucosal barrier is a forerunner of gastric ulcer. Which of the following can both damage the gastric mucosal barrier and stimulate gastric acid secretion?

(A) Bile salts
(B) Epidermal growth factor
(C) Gastrin
(D) *Helicobacter pylori*
(E) Mucous

Answer: D. The discovery of *H. pylori* and its association with peptic ulcer disease, adenocarcinoma, gastric lymphoma, and other diseases make it one of the most significant medical discoveries of this century. In the United States, about 26 million people will suffer from ulcer disease in their lifetime and up to 90% will likely be due to *H. pylori*. *H. pylori* is a gram-negative bacterium with high urease activity, an enzyme that catalyzes the formation of ammonia from urea. The ammonia (NH_3) is converted to ammonium (NH_4^+) in the acid environment of the stomach. The ammonium damages the gastric mucosal barrier because it damages epithelial cells. *H. pylori* also increases gastric acid secretion, possibly by increasing parietal cell mass. This combination of increased acid secretion along with damage to the gastric mucosal barrier promotes the development of gastric ulcer. Bile salts can damage the gastric mucosal barrier, but they do not have a clinically significant effect on acid secretion. Epidermal growth factor, gastrin, and mucous strengthen the gastric mucosal barrier.

94. A 68-year-old woman with hematemesis has hear-burn and stomach pain. Endoscopy shows inflammation involving the gastric body and antrum as well as a small gastric ulcer. Biopsies were positive for *Helicobader pylori*. *H. pylori* damages the gastric mucosa primarily by increasing mucosal levels of which substance?

(A) Ammonium
(B) Bile salts
(C) Gastrin
(D) NSAIDS
(E) Pepsin

Answer: A. *H. pylori* is a bacterium that accounts for 95% of patients with duodenal ulcer and virtually 100% of patients with gastric ulcer when chronic use of aspirin or other non-steroidal anti-inflammatory drugs (NSAIDS) are eliminated. *H. pylori* is characterized by high urease activity, which metabolizes urea to NH_3 (ammonia). Ammonia reacts with H^+ to become ammonium (NH_4^+). This reaction allows the bacterium to withstand the acid environment of the stomach. The ammonium production is believed to be the major cause of cytotoxicity because the ammonium directly damages epithelial cells, increasing the permeability of the gastric mucosal barrier. Bile salts and NSAIDS can also damage the gastric mucosal barrier, but these are not directly related to *H. pylori* infection. Pepsin can exacerbate the mucosal lesions cause by *H. pylori* infection, but pepsin levels are not increased by *H. pylori*. It should be clear that gastrin does not mediate the mucosal damage caused by *H. pylori*.

95. A 62-year-old man with dyspepsia and a history of chronic gastric ulcer has abdominal pain. Endoscopy shows a large ulcer in the proximal gastric body. Biopsies were positive for *Helicnbacter pylori*. Which of the following are used clinically for treatment of gastric ulcers of various etiologies?

	Antibiotics	NSAIDS	H_2 blockers	Proton Pump Inhibitors
(A)	No	No	Yes	Yes
(B)	Yes	No	No	Yes
(C)	Yes	No	Yes	Yes
(D)	Yes	Yes	Yes	Yes
(E)	No	Yes	Yes	Yes

Answer: C. The medical treatment of gastric ulcers is aimed at restoring the balance between acid secretion and mucosal protective factors. Proton pump inhibitors are drugs that covalently bind and irreversibly inhibit the H^+/K^+ adenosine triphosphatase (ATPase) pump, effectively inhibiting acid release. Therapy can also be directed toward histamine release, that is, H_2 blockers, such as cimetidine (Tagamet), ranitidine (Zantac), famotidine (Pepcid), and nizatidine (Axid). These agents selectively block the H_2 receptors in the parietal cells. Antibiotic therapy is used to eradicate the *H. pylori* infection. NSAIDS (nonsteroidal anti-inflammatory agents) can cause damage to the gastric mucosal barrier, which is a forerunner of gastric ulcer.

ZES

96. A patient with Zollinger-Ellison syndrome would be expected to have which of the following changes?

(A) Decreased serum gastrin levels
(B) Increased serum insulin levels
(C) Increased absorption of dietary lipids
(D) Decreased parietal cell mass
(E) Peptic ulcer disease

Answer: E. Zollinger-Ellison syndrome (gastrinoma) is a tumor of the non-β-cell pancreas. The tumor secretes gastrin, which then circulates to the gastric parietal cells to produce increased H^+ secretion, peptic ulcer, and parietal cell growth (trophic effect of gastrin). Because the tumor does not involve the pancreatic β-cells, insulin levels should be unaffected. Absorption of lipids is decreased (not increased) because increased H^+ secretion decreases the pH of the intestinal lumen and inactivates pancreatic lipases.

97. A 57-year-old man is admitted as an emergency for upper GI bleeding. Endoscopy reveals multiple ulcers in the duodenum. Serum gastrin levels are eight-fold higher compared to normal. Zollinger-Ellison syndrome (ZES, gastrinoma) is suspected. Administration of which of the following substances is useful in confirming the diagnosis?

(A) Cholecystokinin (CCK)
(B) Glucose-dependent insulinotropic peptide (GLIP)
(C) Motilin
(D) Pentagastrin
(E) Secretin

Answer: E. ZES is typically caused by gastrin-secreting tumor that is located in the pancreas, duodenal wall, or in lymph nodes. The most simple and reliable test for ZES is secretin injection. Secretin inhibits antral gastrin, but it simulates gastrin secretion in patients with ZES (gastrinoma). Two units of secretin per kilogram body weight are injected intravenously. Serum gastrin levels are measured at various times for 30 min after injection. An increase in serum gastrin of more than 200 ng/mL is diagnostic for ZES. The physiological mechanism of the secretin test remains unclear; however, it is the most important diagnostic test to exclude other conditions associated with increased acid secretion. CCK, GLIP, motilin, and pentagastrin have minimal effects on gastrin secretion and are not diagnostic for gastrinoma.

98. A 53-year-old man with a recurrent history of ulcer disease associated with diarrhea and a strong family history of duodenal ulcer disease is suspected of having Zollinger-Ellison syndrome (gastrinoma). Secretin (2 units/kg) was given as a rapid intravenous injection to test for gastrinoma. Which of the following results would support the existence of gastrinoma following secretin administration?

(A) Decreased serum gastrin
(B) Increased serum gastrin
(C) Inhibition of gastric acid secretion
(D) Inhibition of gastric emptying
(E) Stimulation of pancreatic HCO_3^- secretion

Answer: B. Secretin inhibits gastrin release from the antrum of the stomach, but it stimulates gastrin secretion from a gastrinoma. Thus, patients with a gastrinoma have increased serum gastrin levels within 30 min after secretin administration; whereas secretin decreases serum gastrin levels in normal subjects. Secretin injection is the most simple and reliable test for gastrinoma; however, the physiological mechanism of the secretin test is poorly understood. Secretin normally inhibits gastric acid secretion, but it could conceivably increase acid secretion in patients with gastrinoma because of the increase in gastrin secretion that occurs. Secretin can inhibit gastric emptying when pharmacological doses are given, but this is not diagnostic for gastrinoma. Secretin has a normal physiological effect to stimulate pancreatic HCO_3^- secretion, which is independent of gastrinoma.

Duodenal Ulcer

99. A 42-year-old airline pilot presents to his family physician with a chief complaint of midepigastric pain that is relieved by antacids or eating. Endoscopic evaluation reveals the presence of a duodenal ulcer. Based on the diagnosis, which of the following also would be expected?

(A) Decreased basal acid output
(B) Increased maximal acid output
(C) Decreased gastric emptying of liquids
(D) Decreased gastric emptying of solids
(E) Increased gallbladder emptying

Answer: B. Inflammation of the proximal small intestine results in a decrease in the feedback regulation of gastric function by reducing the input of the enterogastric reflex and enterogastrones to gastric emptying and gastric acid secretion. Absent inhibitory input, basal and maximal acid outputs are increased, and the gastric emptying of liquids and solids is increased.

100. An 84-year-old man with hematemesis and melena is diagnosed with a duodenal ulcer A patient diagnosed with a duodenal ulcer is likely to exhibit which of the following?

	Parietal Cell Density	Acid Secretion	Plasma Gastrin
(A)	Decreased	Decreased	Decreased
(B)	Decreased	Increased	Decreased
(C)	Increased	Decreased	Increased
(D)	Increased	Increased	Decreased
(E)	Increased	Increased	Increased

Answer: D. Duodenal ulcer patients have about 2 billion parietal cells and can secrete about 40 mEq H^+ per hour. Normal individuals have about 50% of these values. Plasma gastrin levels are related inversely to acid secretory capacity because of a feedback mechanism whereby antral acidification inhibits gastrin release. Thus, plasma gastrin levels are reduced in duodenal ulcer patients. Maximal acid secretion and plasma gastrin levels are not diagnostic for duodenal ulcer disease because of significant overlap with the normal population among individuals in each group.

101. A 71-year-old man with hematemesis and melena has a cresenteric ulcer in the duodenum. Lavage dislodged the clot, revealing an underlying raised blood vessel, which was successfully eradicated via cautery with a bipolar gold probe. Which of the following factors are diagnostic for duodenal ulcer?

	Endoscopy	Plasma Gastrin Levels	Rate of Acid Secretion
(A)	No	No	No
(B)	Yes	No	No
(C)	Yes	No	Yes
(D)	Yes	Yes	No
(E)	Yes	Yes	Yes

Answer: B. Neither plasma gastrin levels nor the rate of acid secretion are diagnostic for duodenal ulcer. However, when duodenal ulcer patients are pooled together they exhibit a statistically significant increase in the rate of acid secretion and a statistically significant decrease in plasma gastrin levels. How is this possible? The basal and maximal acid secretion rates of normal subjects range from 1 to 5 mEq/hr and from 6 to 40 mEq/hr, respectively, which overlaps with the basal (2-10 mEq/hr) and maximal (30-80 mEq/hr) acid secretion rates of duodenal ulcer patients. The increase in acid secretion of the average duodenal ulcer patient suppresses the secretion of gastrin from the antrum of the stomach. It should be obvious that endoscopy is diagnostic for duodenal ulcer.

H_2 Blockers

102. Which of the following best describes the pharmacological blockade of histamine H_2 receptors in the gastric mucosa?

(A) It inhibits both gastrin- and acetylcholine-mediated secretion of acid.
(B) It inhibits gastrin-induced but not meal-stimulated secretion of acid.
(C) It has no effect on either gastrin-induced or meal-stimulated secretion of acid.
(D) It prevents activation of adenyl cyclase by gastrin.
(E) It causes an increase in potassium transport by gastric parietal cells.

Answer: C. Histamine (H_2) receptor antagonists inhibit both the direct action of histamine on parietal cells and the potentiating effects of histamine on gastrin-induced and vagal-mediated secretion of acid. Histamine-induced secretion of acid by gastric parietal cells involves stimulation of adenyl cyclase and cyclic AMP-mediated stimulation of the active transport of chloride and potassium-hydrogen ion exchange. ACh (M_3 receptor) and gastrin (CCK_B receptor) activate G_q protein and elevate IP_3/Ca^{2+} causing H^+-K^+-ATPase stimulation.

103. A clinical study is conducted in which gastric acid secretion is stimulated using pentagastrin before and after treatment with a histamine H_2 blocker Which of the following rates of gastric acid secretion (in mEq/hr) is most likely to have occurred in this experiment?

	Pentagastrin Alone	Pentagastrin + H_2 Blocker
(A)	15	15
(B)	25	25
(C)	25	15
(D)	26	28
(E)	40	45

Answer: C. The various secretagogues, which include acetylcholine, gastrin, and histamine, have a multiplicative or synergistic effect on gastric acid secretion. This means that histamine potentiates the effects of gastrin and acetylcholine, and that H_2 blockers attenuate the secretory responses to both acetylcholine and gastrin. Likewise, acetylcholine potentiates the effects of gastrin and histamine, and atropine attenuates the secretory effects of histamine and gastrin. Therefore, in the experiment described, the stimulation of acid secretion by pentagastrin is attenuated by the H_2 blocker because of this multiplicative effect of the secretagogues.

104. A patient with a duodenal ulcer is treated successfully with the drug cimetidine. The basis for cimetidine's inhibition of gastric H^+ secretion is that it…

(A) Blocks muscarinic receptors on parietal cells
(B) Blocks H_2 receptors on parietal cells
(C) Increases intracellular cyclic adenosine monophosphate (cAMP) levels
(D) Blocks H^+-K^+-adenosine triphosphatase (ATPase)
(E) Enhances the action of acetylcholine (ACh) on parietal cells

Answer: B. Cimetidine is a reversible inhibitor of H_2 receptors on parietal cells and blocks H^+ secretion. Cyclic adenosine monophosphate (cAMP) [the second messenger for histamine] levels would be expected to decrease, not increase. Cimetidine also blocks the action of acetylcholine (ACh) to stimulate H^+ secretion. Omeprazole blocks H^+-K^+-adenosine triphosphates (ATPase) directly.

PPI

105. A 47-year-old man takes omeprazole for symptoms of his gastroesophageal reflux disease (GERD). Which of the following best describes the use of proton pump inhibitors (PPI)?

(A) PPIs are not used in the treatment of Zollinger-Ellison syndrome
(B) PPIs inhibit H^+-K^+-ATPase in parietal cells
(C) PPIs are not effective as part of a treatment regimen for Helicobacter pylori
(D) PPIs must be dose-adjusted for renal insufficiency
(E) PPIs decrease gastrin levels

Answer: B. Proton pump inhibitors are used for the treatment of acid-related conditions such as Zollinger-Ellison syndrome. They work by inhibiting the H^+-K^+-ATPase of the gastric parietal cells and cause an increase in gastrin levels. They are used in conjunction with various antibiotics as treatment for the eradication of *H pylori*. They do not need to be dose-adjusted for renal insufficiency.

106. The gastric phase of gastric secretion accounts for about 60% of the acid response to a meal. Which of the following can virtually eliminate the secretion of acid during the gastric phase?

(A) Antiacids (e.g., Rolaids)
(B) Antigastrin antibodies
(C) Atropine
(D) Histamine H_2 blocker
(E) Proton pump inhibitor

Answer: E. A proton pump inhibitor such as omeprazole inhibits all acid secretion by directly inhibiting the H^+, K^+ -ATPase (H^+ pump). The parietal cell has receptors for secretagogues such as gastrin, acetylcholine, and histamine. Therefore, antigastrin antibodies, atropine, and histamine H_2 blockers can reduce the secretion of acid, but none of these can totally eliminate acid secretion. Antacids neutralize gastric acid once it has entered the stomach, but they cannot inhibit acid secretion from parietal cells.

Somatostatin

107. Gastric acid (HCl) secretion is inhibited by…

(A) Somatostatin
(B) Entero-oxyntin
(C) High pH
(D) Amino acids
(E) Acetylcholine (ACh)

Answer: A. Gastric acid (HCl) secretion is inhibited by low pH in the stomach and by somatostatin released from interneurons within the enteric nervous system. Acetylcholine (ACh), the hormone known as entero-oxyntin, and amino acids all stimulate gastric acid secretion.

108. Which of the following stimulus-mediator pairs normally inhibit gastrin release? (CCK, cholecystokinin; GLIP, glucose-dependent insulinotropic peptide)

	Stimulus	Mediator
(A)	Acid	CCK
(B)	Acid	GLIP
(C)	Acid	Somatostatin
(D)	Fatty acid	Motilin
(E)	Fatty acid	Somatostatin

Answer: C. Acid acts directly on somatostatin cells to stimulate the release of somatostatin. The somatostatin decreases acid secretion by directly inhibiting the acid secreting parietal cells and indirectly by inhibiting gastrin secretion from G cells in the antrum. Acid is a weak stimulus for CCK release, but CCK does not inhibit (or stimulate) gastrin release. Acid does stimulate GLIP release. Fatty acids are a weak stimulus for motilin, but motilin does not affect gastrin release. Fatty acids are not thought to stimulate somatostatin release.

109. A patient with alcoholic cirrhosis presents to the emergency room with hematemesis. After stabilizing him with IV fluids, the gastroenterologist administers an agonist/analog of which of the following agents to inhibit gastric acid secretion and visceral blood flow?

(A) Gastrin
(B) Somatostatin
(C) Histamine
(D) Pepsin
(E) Acetylcholine

Answer: B. Somatostatin, located within the gastric antral mucosa, is the principal paracrine secretion involved in the inhibitory feedback of gastric acid secretion by parietal cells. Somatostatin has a short half-life of several minutes, which limits its clinical use. The analog octreotide (Sandostatin), however, can be administered subcutaneously to inhibit the secretion of gastrin and gastric acid and visceral blood flow in patients with bleeding esophageal varices secondary to portal hypertension, after stabilizing with IV fluids, as acute variceal bleeds have a 50% mortality.

VIP

110. Which of the following substances is released from neurons in the GI tract and produces smooth muscle relaxation?

(A) Secretin
(B) Gastrin
(C) Cholecystokinin (CCK)
(D) Vasoactive intestinal peptide (VIP)
(E) Gastric inhibitory peptide (GIP)

Answer: D. Vasoactive intestinal peptide (MP) is a gastrointestinal (GI) neurocrine that causes relaxation of GI smooth muscle. For example, VIP mediates the relaxation response of the lower esophageal sphincter when a bolus of food approaches it, allowing passage of the bolus into the stomach.

Secretin

111. All of the following effects are caused by secretin EXCEPT…

(A) Stimulation of pancreatic bicarbonate (HCO_3^-) secretion
(B) Enhancement of bile acid secretion
(C) Potentiation of pancreatic enzyme secretion by cholecystokinin (CCK)
(D) Inhibition of gastric muscle contraction

Answer: B. Although secretin regulates the volume of biliary secretion, it neither enhances nor inhibits the secretion of bile salts. Secretin's major effect is to stimulate bicarbonate (HCO_3^-) secretion by the pancreas and the liver. In addition, it potentiates the effect of cholecystokinin (CCK) on pancreatic enzyme secretion. Secretin also diminishes gastric emptying, probably by directly reducing gastric smooth muscle contractility.

112. Which of the following will inhibit stomach contractions?

(A) Acetylcholine (ACh)
(B) Motilin
(C) Gastrin
(D) Secretin
(E) Histamine

Answer: D. Secretin has a direct inhibitory effect on the smooth muscle fibers forming the stomach wall. Acetylcholine (ACh) and motilin, and possibly gastrin, increase the force of stomach contractions. Histamine has no direct effect on stomach contractions.

113. Intestinal motility is increased by all of the following EXCEPT…

(A) Cholecystokinin (CCK)
(B) Secretin
(C) Gastrin
(D) Insulin
(E) Motilin

Answer: B. Secretin decreases intestinal and gastric contractile force. Cholecystokinin (CCK), gastrin, insulin, and motilin all increase intestinal contractions.

114. The major stimulus for the release of secretin is…

(A) Protein digestion products
(B) Histamine
(C) Somatostatin
(D) Hydrochloric acid (HCl)
(E) Cholecystokinin (CCK)

Answer: D. Secretin and cholecystokinin (CCK) are hormones released from endocrine cells located in the proximal intestine. Although the release of both hormones is stimulated by the presence of chyme in the small intestine, it is the low pH resulting from hydrochloric acid (HCl) in the chyme that is the major stimulus for the release of secretin. Protein digestion products (e.g., amino acids, particularly phenylalanine) stimulate both secretin and CCK secretion, but are the major stimuli for CCK secretion. CCK potentiates the effects of secretin but does not affect its release. Somatostatin and histamine have no effect on CCK or secretin secretion.

115. Major stimulant of HCO_3^- secretion in bile duct.

(A) Gastrin
(B) Secretin
(C) Cholecystokinin
(D) Gastric inhibitory peptide

Answer: B. Secretin as a stimulant for bicarbonate secretion is mainly via the pancreas, but it does have a slight effect in increasing the bicarbonate level of bile.

116. A 33-year-old woman who has been taking large doses of NSAIDs for her menstrual cramps presents with burning epigastric pain. The pain improves after eating a meal, suggesting a gastric versus duodenal ulcer. In the case of a duodenal ulcer, which of the following is the major factor that protects the duodenal mucosa from damage by gastric acid?

(A) Pancreatic bicarbonate secretion
(B) The endogenous mucosal barrier of the duodenum
(C) Duodenal bicarbonate secretion
(D) Hepatic bicarbonate secretion
(E) Bicarbonate contained in bile

Answer: A. Pancreatic bicarbonate secretion into the small intestine is essential for neutralization of gastric acid emptied into the small intestine. Unlike the gastric mucosal lining, the mucosal surface of the small intestine does not provide a significant endogenous defense mechanism against the insult of HCl. Upon delivery into the proximal small intestine, hydrogen ions stimulate the release of the hormone secretin from the intestinal wall, which in turn stimulates pancreatic bicarbonate secretion. In fact, the acid output of the stomach during a meal is matched equally by the pancreatic output of bicarbonate. Although the liver secretes bicarbonate and bile contains bicarbonate, the amounts are not sufficient for acid neutralization.

117. Which of the following combination of agents will produce the highest rate of pancreatic bicarbonate secretion?

(A) Secretin plus histamine
(B) Cholecystokinin plus acetylcholine
(C) Gastrin plus vagal stimulation
(D) Secretin plus cholecystokinin
(E) Cholecystokinin plus phenylalanine

Answer: D. All will lead to bicarbonate secretion, but choice D is more direct.

CCK

118. A secretion involved in digestion is collected as it enters the gastrointestinal tract. Secretin increases the pH, and cholecystokinin increases the volume, of this secretion. Once in the GI tract, this secretion contributes to the digestion of proteins. Which of the following most likely represents the secretion?

(A) Saliva
(B) Gastric juice
(C) Bile
(D) Pancreatic juice
(E) Trypsin

Answer: D. Main function of CCK is to stimulate pancreatic secretion. Main function of secretin is to stimulate pancreatic bicarbonate secretion. Both inhibit/reduce gastric juice secretion. Bile and trypsin (as trypsinogen) secretions are stimulated by CCK.

119. Which stimulus is most important for the regulation of pancreatic enzyme secretion?

(A) Secretin
(B) Histamine
(C) Cholecystokinin (CCK)
(D) Bombesin
(E) Motilin

Answer: C. Cholecystokinin (CCK) and secretin both stimulate the pancreas. CCK, however, is responsible for the secretion of pancreatic enzymes.

120. Which of the following is true about the secretion from the exocrine pancreas?

(A) It has a higher Cl^- concentration than does plasma
(B) It is stimulated by the presence of HCO_3^- in the duodenum
(C) Pancreatic HCO_3^- secretion is increased by gastrin
(D) Pancreatic enzyme secretion is increased by cholecystokinin (CCK)
(E) It is hypotonic

Answer: D. The major action in pancreatic secretion is HCO_3^- (which is found in higher concentration than in plasma), and the Cl^- concentration is lower than in plasma. Pancreatic secretion is stimulated by the presence of fatty acids in the duodenum. Secretin (not gastrin) stimulates pancreatic HCO_3^- secretion, and cholecystokinin (CCK) stimulates pancreatic enzyme secretion. Pancreatic secretions are always isotonic regardless of flow rate.

121. Polypeptide that primarily stimulates release of pancreatic amylase.

(A) Gastrin
(B) Secretin
(C) Cholecystokinin
(D) Gastric inhibitory peptide

Answer: C. The peptide hormone that stimulates the secretion of pancreatic enzymes is cholecystokinin.

122. All of the following stimulate cholecystokinin (CCK) secretion EXCEPT…

(A) Amino acids
(B) Fatty acids
(C) Gastric acids
(D) Bile acids

Answer: D. Amino acids, particularly phenylalanine, fatty acids, and monoglycerides all directly stimulate the release of cholecystokinin (CCK) from intestinal endocrine cells. Hydrochloric acid (HCl), which lowers the pH of chyme entering the intestine, also causes CCK secretion. Bile acids do not influence CCK secretion.

123. Cholecystokinin (CCK) inhibits…

(A) Gastric emptying
(B) Pancreatic HCO_3^- secretion
(C) Pancreatic enzyme secretion
(D) Contraction of the gallbladder
(E) Relaxation of the sphincter of Oddi

Answer: A. Cholecystokinin (CCK) inhibits gastric emptying and therefore helps to slow the delivery of food from the stomach to the intestine during periods of high digestive activity. CCK stimulates both functions of the exocrine pancreas HCO_3^-, secretion and digestive enzyme secretion. It also stimulates the delivery of bile from the gallbladder to the small intestinal lumen by causing contraction of the gallbladder while relaxing the sphincter of Oddi.

124. Which of the following substances inhibits gastric emptying?

(A) Secretin
(B) Gastrin
(C) Cholecystokinin (CCK)
(D) Vasoactive intestinal peptide (VIP)
(E) Gastric inhibitory peptide (GIP)

Answer: C. Cholecystokinin (CCK) is the most important hormone for digestion and absorption of dietary fat. In addition to causing contraction of the gallbladder, it inhibits gastric emptying. As a result, chyme moves more slowly from the stomach to the small intestine, thus allowing more time for fat digestion and absorption.

125. Which of the following hormones is released by the presence of fat and protein in the small intestine and has a major effect to decrease gastric emptying?

(A) Cholecystokinin (CCK)
(B) Glucose-dependent insulinotropic peptide (GLIP)
(C) Gastrin
(D) Motilin
(E) Secretin

Answer: A. CCK is the only gastrointestinal hormone that inhibits gastric emptying under physiological conditions. This inhibition of gastric emptying keeps the stomach full for a prolonged time, which is one reason why a breakfast containing fat and protein "sticks with you" better than breakfast meals containing mostly carbohydrates. CCK also has a direct effect on the feeding centers of the brain to reduce further eating. Although CCK is the only gastrointestinal hormone that inhibits gastric emptying, all of the gastrointestinal hormones with the exception of gastrin are released to some extent by the presence of fat in the intestine.

126. Which stimulus is most important for the regulation of gallbladder emptying?

(A) Secretin
(B) Histamine
(C) Cholecystokinin (CCK)
(D) Bombesin
(E) Motilin

Answer: C. CCK is responsible for stimulating the gallbladder to contract.

127. Causes contraction of the gall bladder.

(A) Gastrin
(B) Secretin
(C) Cholecystokinin
(D) Gastric inhibitory peptide

Answer: C. Contraction of the gall bladder is caused by cholecystokinin.

128. Which of the following is the primary physiological stimulus of gallbladder contraction in the digestive period?

(A) Acid-induced release of secretin from the smal1 intestine
(B) Fat-induced release of cholecystokinin from the small intestine
(C) Acid-induced release of motilin from the small intestine
(D) Distension-induced release of glucagon from the small intestine
(E) Amino acid-induced release of motilin from the small intestine

Answer: B. The delivery of food into the small intestine is characterized by prompt emptying of the gallbladder, resulting from fat-induced release of cholecystokinin. Secretin stimulates pancreatic bicarbonate secretion. Glucagon is involved in nutrient metabolism and motilin is an interdigestive hormone responsible for migrating motor complex activity.

129. A 42-year-old woman presents to the ER with right upper quadrant pain that developed after eating dinner. She is diagnosed as having cholecystitis. Which of the following would be expected with contraction of the gallbladder following a meal?

(A) It is inhibited by a fat-rich meal.
(B) It is inhibited by the presence of amino acids in the duodenum.
(C) It is stimulated by atropine.
(D) It occurs in response to cholecystokinin.
(E) It occurs simultaneously with the contraction of the sphincter of Oddi.

Answer: D. Cholecystokinin (CCK) is released from the upper small intestine in response to partially hydrolyzed dietary lipids and proteins and promotes gallbladder emptying. Gallbladder contraction and sphincter of Oddi relaxation are necessary for delivery of bile into the duodenum. These muscular actions are under both hormonal and neural control. CCK contracts gallbladder smooth muscle by a direct action on the muscle and through activation of vagal afferent fibers leading to a vago-vagal reflex. Relaxation of sphincter of Oddi smooth muscle occurs via activation of inhibitory enteric nerves. Gallbladder contraction is also promoted by vagal stimulation, which is cholinergically mediated and blocked by the muscarinic receptor antagonist, atropine.

130. Cholecystokinin is important in fat absorption because it:

(A) Stimulates the gall bladder to contract
(B) Potentiates effect of secretin on the secretion of bicarbonate by the pancreas
(C) Stimulates the secretion of pancreatic lipase
(D) Increases the strength of gastric peristalsis

Answer: A, B, and C. Cholecystokinin has 2 important roles in the absorption of fat. It causes contraction of the gall bladder and relaxation of the sphincter of Oddi, permitting bile to enter the duodenum. It is also responsible for the secretion of pancreatic enzymes, which would include pancreatic lipases. Cholecystokinin and secretin potentiate each other's effect on the pancreas (synergistic action). Cholecystokinin has a small effect on stomach motility, but it is debatable whether this is significant.

131. Which of the following occurs during the phase of the digestive period when plasma cholecystokinin levels are maximal?

(A) pH of hepatic bile increases
(B) Secretion of HCl by parietal cells increases
(C) Rate of stomach emptying increases
(D) Motility of the colon is minimal
(E) Volume of the gallbladder increases

Answer: A. CCK also increases hepatic bile production as secretin does. All other choices are opposite of CCK function.

132. Cholecystokinin (CCK) has some gastrin-like properties because both CCK and gastrin

(A) Are released from G cells in the stomach
(B) Are released from I cells in the duodenum
(C) Are members of the secretin homologous family
(D) Have five identical C-terminal amino acids
(E) Have 90% homology of their amino acids

Answer: D. The two hormones have five identical amino acids at the C terminus. Biologic activity of cholecystokinin (CCK) is associated with the seven C-terminal amino acids, and biologic activity of gastrin is associated with the four C-terminal amino acids. Because this CCK heptapeptide contains the five common amino acids, it is logical that CCK should have some gastrin-like properties. G cells secrete gastrin. I cells secrete CCK. The secretin family includes glucagon.

133. Which of the following competes for many of the same receptor sites as gastrin because of a chemical structure similar to gastrin?

(A) Secretin
(B) Glucagon
(C) Histamine
(D) Cholecystokinin
(E) Gastric inhibitory peptide

Answer: D. Gastrin and cholecystokinin are both peptides with similar structures.

134. Cholecystokinin (CCK) and gastrin share multiple effects at pharmacological concentrations. Which of the following effects do CCK and gastrin share (or not share) at physiological concentrations?

	Stimulation of Acid Secretion	Inhibition of Gastric Emptying	Stimulation of Gastric Mucosal Growth	Stimulation of Pancreatic Growth
(A)	Not shared	Not shared	Not shared	Not shared
(B)	Not shared	Not shared	Shared	Not shared
(C)	Not shared	Shared	Not shared	Not shared
(D)	Shared	Shared	Not shared	Not shared
(E)	Shared	Shared	Shared	Shared

Answer: A. Gastrin and CCK do not share any effects on gastrointestinal function at normal, physiological conditions; however, they have identical actions on gastrointestinal function when pharmacological doses are administered. Gastrin stimulates gastric acid secretion and mucosal growth throughout the stomach and intestines under physiological conditions. CCK stimulates growth of the exocrine pancreas and inhibits gastric emptying under normal conditions. CCK also stimulates gallbladder contraction, relaxation of the sphincter of Oddi, and secretion of bicarbonate and enzymes from the exocrine pancreas.

Pancreas

135. After secretion into the duodenum, the enzyme trypsinogen is converted into its active form, trypsin, by:

(A) Enterokinase
(B) Procarboxypeptidase
(C) Lysolecithin
(D) An alkaline pH
(E) An acid pH

Answer: A. Enterokinase must be present in the duodenum specifically to activate trypsinogen to trypsin. Trypsin then activates the remaining pancreatic proteases. Without enterokinase, pancreatic proteases will not be adequately activated and protein digestion cannot proceed normally.

136. A 38-year-old man has dinner one evening at his favorite steakhouse. Several hours later, the chyme reaches the duodenum. After secretion of trypsinogen into the duodenum, the enzyme is converted into its active form, trypsin, by which of the following?

(A) Enteropeptidase
(B) Procarboxypeptidase
(C) Pancreatic lipase
(D) Chymotrypsin
(E) An alkaline pH

Answer: A. Liberation of the enzyme enteropeptidase (enterokinase) from the duodenal mucosal cells causes the inactive trypsinogen to be converted to the active form, trypsin. Enteropeptidase contains 41% polysaccharide. It is this high level of polysaccharide that protects enteropeptidase from digestion. Trypsin is responsible for the conversion of chymotrypsinogens and other proenzymes into their active forms.

137. Various proteolytic enzymes are secreted in an inactive form into the lumen of the gastrointestinal tract. Which of the following substances is/are important for activating one or more proteolytic enzymes, converting them to an active form?

	Trypsin	Enterokinase	Pepsin
(A)	No	No	No
(B)	No	No	Yes
(C)	No	Yes	No
(D)	Yes	Yes	No
(E)	Yes	Yes	Yes

Answer: E. Essentially all proteolytic enzymes are secreted in an inactive form, which prevents autodigestion of the secreting organ. Enterokinase is physically attached to the brush border of the enterocytes which line the inner surface of the small intestine. Enterokinase activates trypsinogen to become trypsin in the gut lumen. The trypsin then catalyzes the formation of additional trypsin from trypsinogen as well as several other proenzymes (e.g., chymotrypsinogen, procarboxypeptidase, proelastase, and others). Pepsin is first secreted as pepsinogen, which has no proteolytic activity. However, as soon as it comes into contact with hydrochloric acid, and especially in contact with previously formed pepsin plus hydrochloride acid, it is activated to form pepsin.

138. A patient deficient in trypsin inhibitor has primary abnormality that involves the:

(A) Oxyntic cells of the stomach
(B) Pancreas
(C) Small intestine epithelium
(D) Salivary glands
(E) Liver

Answer: B. Trypsin inhibitor is a substance secreted by the pancreas along with the protein proenzymes.

139. Cystic fibrosis is the most common cause of pancreatitis in children. Which of the following best explains the mechanism of cystic fibrosis-induced pancreatitis?

(A) Activation of enterokinase
(B) Activation of trypsin inhibitor
(C) Autodigestion of pancreas
(D) Excessive secretion of CCK
(E) Gallstone obstruction

Answer: C. Pancreatitis is inflammation of the pancreas. The pancreas secretes digestive enzymes into the small intestine that are essential in the digestion of fats, proteins, and carbohydrates. Reduced secretion of fluid into the pancreatic ducts in cystic fibrosis cause these digestive enzymes to accumulate in the ducts. The digestive enzymes then become activated in the pancreatic ducts (which typically would not occur) and can begin to "digest" the pancreas, leading to inflammation and a myriad of other problems (cysts and internal bleeding). Enterokinase is located at the brush border of intestinal enterocytes where it normally activates trypsin from its precursor, trypsinogen. Trypsin inhibitor is normally present in the pancreatic ducts where it prevents trypsin from being activated, and thus prevents autodigestion of the pancreas. When the ducts are blocked in cystic fibrosis, the available trypsin inhibitor is insufficient to prevent trypsin from being activated. Excessive secretion of CCK does not occur in cystic fibrosis. Gallstone obstruction can lead to pancreatitis (by autodigestion) when the obstruction prevents pancreatic juice from entering the intestine, but this is unrelated to cystic fibrosis.

140. A 42-year-old man with alcoholism presents to the ER with epigastric abdominal pain associated with nausea and vomiting. He is diagnosed with acute pancreatitis. Which of the following best describes pancreatic function?

(A) Secretin inhibits HCO_3^- secretion from the pancreas
(B) Pancreatic lipase converts triglycerides to monoglycerides and free fatty acids
(C) Serum amylase would be decreased in this patient
(D) Serum lipase would be decreased in this patient
(E) Hypertriglyceridemia is not associated with pancreatitis

Answer: B. Pancreatic lipase converts triglycerides to monoglycerides and free fatty acids. Secretin stimulates bicarbonate secretion from the pancreas. Serum amylase and lipase would be expected to be increased in a patient presenting with acute pancreatitis. Gallstones are the most common cause of pancreatitis. Alcohol, hypertriglyceridemia, drugs, and endoscopic retrograde cholangiopancreatography (ERCP) are other common causes.

141. Trypsin and chymotrypsin are similar in that they:

(A) Are activated in the intestine
(B) Occur in precursor form in the pancreas
(C) Are active in an alkaline or neutral medium
(D) Require vitamin K for their synthesis

Answer: A, B, and C. These are proteases that are converted to their active forms in the duodenum (initiated by enterokinase acting on trypsinogen). The inactive precursors are secreted by the pancreas. These enzymes are active in the neutral or slightly alkaline environment of the small intestine.

Neutralization in Duodenum

142. Neutralization of acidic stomach contents entering duodenum is due to:

(A) Pancreatic juice
(B) Bile
(C) Secretions of intestinal glands
(D) Saliva

Answer: A, B, and C. The main mechanism that neutralizes stomach acid entering the duodenum is the release of secretin that causes the release of pancreatic bicarbonate. Answer A would be the best answer to this question. There is, however, some contribution from bile and intestinal secretions. There is no contribution from salivary secretions whose bicarbonate will be neutralized by stomach acid.

143. The presence of acid chyme in the duodenum will lead to:

(A) Increased pepsin production
(B) Increased gastric emptying
(C) Increased gastric acid production
(D) Increased HCO_3^- secretion by biliary ducts
(E) Relaxation of the gallbladder

Answer: D. The presence of acid and amino acids in the duodenum stimulates secretion of secretin, and increases bicarbonate secretion by the pancreatic and biliary system. Secretin also inhibits acid production by the stomach. The products of fat and protein digestion in the small intestine stimulate CCK secretion and cause increased pancreatic secretion, contraction of the gallbladder, promotion of intestinal motility, and inhibition of gastric emptying.

Bile and Gall Bladder

144. Bile contains:

(A) Lecithin
(B) Cholesterol
(C) Micelles
(D) Chymotrypsinogen

Answer: A, B, and C. Micelles, which are composed of bile salts, are also a vehicle for lipid transport and contain lipids such as lecithin and cholesterol. Chymotrypsinogen is the precursor to a protease secreted by the pancreas.

145. During storage in gall bladder, bile undergoes an increase in:

(A) Osmotic pressure
(B) H^+ concentration
(C) Water content
(D) Bile salt concentration

Answer: D. The main change in the gall bladder is the absorption of water and electrolytes.

Bile salts and bile pigments are not absorbed, but the removal of water causes these substances to become more concentrated. To protect the lining epithelium, the gallbladder secretes mucins and H^+ to neutralize the alkaline bile. (It is like HCO_3^- in the stomach, not actually changing the pH of the content.)

146. The major factor controlling the secretion of bile salts from the liver is the amount of…

(A) Secretin released during a meal
(B) Fat entering the small intestine
(C) Bile acids produced by the liver
(D) Bile reabsorbed from the intestine
(E) Cholecystokinin (CCK) released during a meal

Answer: D. The amount of bile synthesized each day is not sufficient to absorb all of the fat digested. However, because the bile salts are absorbed by the intestine and returned to the liver, they can be used over and over again. The amount of bile secreted each day is thus proportional to the amount absorbed from the intestine.

147. Which of the following levels is the primary determinant of the rate of bile acid secretion by the hepatocytes?

(A) Plasma levels of cholecystokinin
(B) Plasma levels of secretin
(C) Plasma levels of bile acids
(D) Level of parasympathetic nervous activity
(E) Level of sympathetic nervous activity

Answer: C. High delivery rates in sinusoidal blood cause stimulation of bile fluid secretion, but inhibits synthesis of new bile acids.

148. Which of the following best describes bile acid function?

(A) They are essentially water insoluble
(B) The majority of bile acids are absorbed by passive diffusion
(C) Glycine conjugates are more soluble than taurine conjugates
(D) The amount lost in the stool each day represents the daily loss of cholesterol
(E) The bile acid-dependent fraction of bile is stimulated by the hormone secretin

Answer: D. Although only small amounts of bile acids are lost in the stool each day, the loss represents the only route of elimination of cholesterol from the body.

The predominant organic component of bile is the bile salts, which make up about 67% of the total solutes. Bile salts are amphiphilic molecules, that is, they exhibit both water and lipid solubility. Primary bile acids, cholic acid, and chenodeoxycholic acid are synthesized from cholesterol. Secondary bile acids, deoxycholic acid, and lithocholic acid are produced by biotransformation of primary bile acids by intestinal bacteria. Prior to secretion, the bile acids are conjugated with either glycine or taurine, which greatly enhances their water solubility. In general, taurine conjugates are more water-soluble than glycine conjugates. The majority of bile acid is absorbed via a Na^+-K^{+-}ATPase in the terminal ileum.

149. Where does the absorption of bile acids occur?

(A) Fundus of the stomach
(B) Antrum of the stomach
(C) Duodenum of the intestine
(D) Ileum of the intestine
(E) Colon

Answer: D. Bile acids are absorbed in the terminal ileum by a Na^+-dependent cotransport process.

150. Which of the following is the site of Na^+-bile acid cotransport?

(A) Gastric antrum
(B) Gastric fundus
(C) Duodenum
(D) Ileum
(E) Colon

Answer: D. Bile salts are recirculated to the liver in the enterohepatic circulation via a Na^+-bile acid cotransporter located in the ileum of the small intestine.

151. Secondary bile acids are formed…

(A) In the liver from cholesterol
(B) By the conjugation of bile acids with taurine or glycine
(C) Both
(D) Neither

Answer: D. Secondary bile acids are formed in the intestine by the deconjugation and dehydroxylation of primary bile acids. Primary bile acids are synthesized from cholesterol in the liver and then conjugated with taurine or glycine to form primary bile salts.

152. Unconjugated bilirubin is:

(A) Necessary for intestinal absorption of fatty acids
(B) Highly soluble in plasma
(C) The form of bilirubin normally found in the bile canaliculus
(D) Elevated in the serum in case of hemolytic jaundice

Answer: D. Unconjugated bilirubin is a lipid-soluble compound produced from the degradation of hemoglobin. It is transported to the liver attached to protein because it is lipid rather than water soluble. In the liver it is made water soluble and excreted in the bile. Bile is important for assimilation of dietary lipid (~50%), and unconjugated bilirubin is less water soluble and thus it is not readily incorporated in the bile.

153. An 18-year-old woman decides to get a tattoo for her birthday. Two months later she presents with a fever, right upper quadrant pain, nausea vomiting, and jaundice. Which of the following laboratory values would most likely be found in a patient with infectious hepatitis?

(A) An increase in plasma alkaline phosphatase
(B) An increase in plasma bile acids
(C) An increase in both direct and indirect plasma bilirubin
(D) A decrease in plasma alkaline phosphatase
(E) A decrease in both direct and indirect plasma bilirubin

Answer: C. Infectious hepatitis is a systemic infection predominantly affecting the liver. When jaundice appears, serum bilirubin rises, and, in most instances, total bilirubin is equally divided between the conjugated (direct) and unconjugated (indirect) fractions. The bilirubin in serum represents a balance between input from production of bilirubin and hepatic/biliary removal of the pigment. Hyperbilirubinemia may result from (1) overproduction of bilirubin; (2) impaired uptake, conjugation, or removal of bilirubin; or (3) regurgitation of unconjugated or conjugated bilirubin from damaged hepatocytes or bile ducts. Alkaline phosphatase, which is excreted in bile, increases in patients with jaundice due to bile duct obstruction, but generally normal when the jaundice is due to acute hepatocellular disease. Bile acids are synthesized in the liver by a series of enzymatic steps that also involve cholesterol catabolism. Liver disease decreases bile acid synthesis.

154. A 43-year-old obese woman with a history of gallstones is admitted to the emergency department because of excruciating pain in the upper right quadrant. The woman is jaundiced and x-ray suggests obstruction of the common bile duct. Which of the following values of direct and indirect bilirubin are most likely to be present in the plasma of this woman (in milligrams per deciliter)?

	Direct	**Indirect**
(A)	1.0	1.3
(B)	2.3	2.4
(C)	5.0	1.7
(D)	1.8	6.4
(E)	6.8	7.5

Answer: C. About 20% of persons older than 65 years have gallstones (cholelithiasis) in the United States, and 1 million newly diagnosed cases of gallstones are reported each year. Gallstones are the most common cause of biliary obstruction. Regardless of the cause of gallstones, serum bilirubin values (especially direct or conjugated) are usually elevated. Indirect or unconjugated bilirubin values are usually normal or only slightly elevated. Only answer C shows a high level of direct bilirubin (conjugated bilirubin) compared to the level of indirect bilirubin (unconjugated bilirubin).

155. Obstruction of bile duct would result in:

(A) Increased content of lipids in feces
(B) Increased excretion of dietary cholesterol in feces
(C) Increased conjugated bile pigments in the serum
(D) Decreased excretion of bilirubin glucuronides in urine

Answer: A and C. Obstruction of the bile duct means bile cannot enter the duodenum. Bile micelles are important in the digestion and absorption of lipid, and their absence causes increased lipids in the stool. Also, conjugated (water-soluble) bile pigments will appear in the plasma, and because they are water-soluble they will be filtered by the kidney and appear in the urine.

156. A 47-year-old woman presents to her primary care physician with jaundice. She is found to have elevated levels of direct (conjugated) plasma bilirubin. Which of the following is the most likely diagnosis?

(A) Gilbert syndrome
(B) Crigler-Najjar syndrome type I
(C) Crigler-Najjar syndrome type II
(D) Obstruction of the common bile duct
(E) Hemolytic anemia

Answer: D. When plasma bilirubin is increased due to bile duct obstruction, it is generally the conjugated form of bilirubin that increases due to reabsorption of bilirubin glucuronide into the blood. All other choices are found in conditions with an increase in the indirect (unconjugated) form of bilirubin.

GIP/GLIP

157. Which of the following substances is secreted in response to an oral glucose load?

(A) Secretin
(B) Gastrin
(C) Cholecystokinin (CCK)
(D) Vasoactive intestinal peptide (VIP)
(E) Glucose-dependent insulinotropic peptide (GIP)

Answer: E. Glucose-dependent insulinotropic peptide (GIP) is the only gastrointestinal (GI) hormone that is released in response to all three categories of nutrients-fat, protein, and carbohydrate. Oral glucose releases GIP, which, in turn, causes the release of insulin from the endocrine pancreases this action of GIP explains why oral glucose is more effective than intravenous glucose in releasing insulin.

158. A 10-year-old boy consumes a cheeseburger, fries, and chocolate shake. The meal stimulates the release of several gastrointestinal hormones. Which of following hormones release from duodenal mucosa can be stimulated by the presence of fat, carbohydrate, or protein in the duodenum?

(A) Cholecystokinin (CCK)
(B) Glucose-dependent insulinotropic peptide (GLIP)
(C) Gastrin
(D) Motilin
(E) Secretin

Answer: B. GLIP is the only gastrointestinal hormone released by all three major foodstuffs (fats, proteins, and carbohydrates). The presence of fat and protein in the small intestine stimulates the release of CCK, but carbohydrates do not stimulate its release. The presence of protein in the antrum of the stomach stimulates the release of gastrin, but fat and carbohydrates do not stimulate its release. Fat has a minor effect to stimulate the release of motilin and secretin, but neither hormone is released by the presence of protein or carbohydrate in the gastrointestinal tract.

159. A 23-year-old medical student consumes a cheeseburger, fries, and chocolate shake. Which of the following hormones produce physiological effects at some point over the next several hours?

	Gastrin	Secretin	Cholecystokinin	GLIP
(A)	No	Yes	Yes	Yes
(B)	Yes	No	Yes	Yes
(C)	Yes	Yes	No	Yes
(D)	Yes	Yes	Yes	Yes
(E)	Yes	Yes	Yes	Yes

Answer: D and E. All of the GI hormones are released following a meal and all have physiological effects.

160. A clinical experiment is conducted in which one group of subjects is given 50 g of glucose intravenously and another group is given 50 g of glucose orally. Which of the following factors can explain why the oral glucose load is cleared from the blood at a faster rate compared to the intravenous glucose load? (CCK, cholecystokinin; GLIP, glucose-dependent insulinotropic peptide; VIP, vasoactive intestinal peptide)

(A) CCK-induced insulin release
(B) CCK-induced VIP release
(C) GLIP-induced glucagon release
(D) GLIP-induced insulin release
(E) VIP-induced GLIP release

Answer: D. GLIP is released by the presence of fat, carbohydrate, or protein in the gastrointestinal tract. GLIP is a strong stimulator of insulin release and is responsible for the observation that an oral glucose load releases more insulin and is metabolized more rapidly than an equal amount of glucose administered intravenously. Intravenously administered glucose does not stimulate the release of GLIP. Neither CCK nor VIP stimulates the release of insulin. GLIP does not stimulate glucagon release, and glucagon has the opposite effect of insulin, that is, it would decrease the rate of glucose clearance from the blood. VIP does not stimulate GLIP release.

GLP-1

161. A 52-year-old woman who has been dieting for several weeks breaks down and eats half a pan of frosted brownies. Insulin secretion following a carbohydrate-rich meal is stimulated by which of the following?

(A) Gastrin
(B) CCK
(C) Serotonin
(D) VIP
(E) GLP-1

Answer: E. Glucagon-like polypeptide 1 (GLP-1) is a product of glucagon metabolism in the L cells of the lower intestinal tract. GLP-1 has no definite biological activity by itself but is processed further by removal of its amino-terminal amino acid residues, and the product, GLP-1 (7-36), is a potent stimulator of insulin secretion that also increases glucose utilization.

Hormones Interactions

162. Gastric acid is secreted when a meal is consumed. Which of the following factors have a direct action on the parietal cell to stimulate acid secretion?

	Gastrin	Somatostatin	Acetylcholine	Histamine
(A)	No	No	Yes	Yes
(B)	Yes	No	No	Yes
(C)	Yes	No	Yes	Yes
(D)	Yes	Yes	Yes	Yes
(E)	Yes	Yes	No	Yes

Answer: C. Gastrin, acetylcholine, and histamine can directly stimulate parietal cells to secrete acid. These three secretagogues also have a multiplicative effect on acid secretion such that inhibition of one secretagogue reduces the effectiveness of the remaining two secretagogues. Acetylcholine also has an indirect effect to increase acid secretion by stimulating gastrin secretion from G cells. Somatostatin inhibits acid secretion.

163. The gastrointestinal hormones have physiological effects that can be elicited at normal concentrations as well as pharmacological effects that require higher than normal concentrations. What is the direct physiological effect of the various hormones on gastric acid secretion? (GLIP, glucose-dependent insulinotropic peptide)

	Gastrin	Secretin	Cholecystokinin	GLIP	Motilin
(A)	No effect	Stimulate	Stimulate	No effect	No effect
(B)	Stimulate	Inhibit	No effect	Inhibit	No effect
(C)	Stimulate	Inhibit	No effect	No effect	No effect
(D)	Stimulate	Inhibit	Inhibit	Stimulate	Stimulate
(E)	Stimulate	Stimulate	Inhibit	Inhibit	No effect

Answer: B. Gastrin stimulates gastric acid secretion, and secretin and GLIP inhibit gastric acid secretion under normal, physiological conditions. It is important to differentiate the physiological effects of the gastrointestinal hormones from their pharmacological actions. For example, gastrin and cholecystokinin (CCK) have identical actions on gastrointestinal function when large, pharmacological doses are administered, but they do not share any actions at normal, physiological concentrations. Likewise, GLIP and secretin share multiple actions when pharmacological doses are administered, but only one action is shared at physiological concentrations: inhibition of gastric acid secretion.

164. Which of the following factors can inhibit gastric acid secretion? (GLIP, glucose-dependent insulinotropic peptide)

	Somatostatin	Secretin	GLIP	Enterogastrones	Nervous reflexes
(A)	No	No	Yes	No	Yes
(B)	No	Yes	No	No	No
(C)	No	Yes	No	Yes	No
(D)	Yes	No	No	Yes	Yes
(E)	Yes	No	Yes	No	No
(F)	Yes	Yes	Yes	Yes	Yes

Answer: F. All of these factors can inhibit gastric acid secretion under normal physiological conditions. Gastric acid stimulates the release of somatostatin (a paracrine factor), which has a direct effect on the parietal cell to inhibit acid secretion as well as an indirect effect mediated by suppression of gastrin secretion. Secretin and GLIP inhibit acid secretion through a direct action on parietal cells as well as indirectly through suppression of gastrin secretion. Enterogastrones are unidentified substances released from the duodenum and jejunum that directly inhibit acid secretion. When acid or hypertonic solutions enter the duodenum, a neurally mediated decrease in gastric acid secretion follows.

165. Which of the following factors have a physiologic role to stimulate the release of hormones or stimulate nervous reflexes, which in turn can inhibit gastric acid secretion?

	Acid	Fatty acids	Hyperosmotic solutions	Isotonic solutions
(A)	No	No	Yes	No
(B)	No	No	Yes	Yes
(C)	Yes	Yes	No	Yes
(D)	Yes	Yes	Yes	Yes
(E)	Yes	Yes	Yes	No

Answer: E. The presence of acid, fatty acids, and hyperosmotic solutions in the duodenum and jejunum lead to suppression of acid secretion through a variety of mechanisms. Acid stimulates the secretion of secretin from the small intestine, which in turn inhibits acid secretion from parietal cells. Acidification of the antrum and oxyntic gland area of the stomach stimulates the release of somatostatin, which in turn inhibits acid secretion by a direct action on the parietal cells and an indirect action mediated by suppression of gastrin secretion. The presence of fatty acids in the small intestine stimulates the release of GLIP (glucose-dependent insulinotropic peptide), which inhibits acid secretion both directly (parietal cell inhibition) and indirectly (by decreasing gastrin secretion). Hyperosmotic solutions in the small intestine cause the release of unidentified enterogastrones, which directly inhibit acid secretion from parietal cells. Isotonic solutions have no effect on acid secretion.

166. One of the following hormones can stimulate growth of the intestinal mucosa and two other hormones can stimulate pancreatic growth. Which three hormones are these?

	Gastrin	Secretin	Cholecystokinin	GLIP	Motilin
(A)	No	Yes	Yes	Yes	No
(B)	Yes	No	Yes	No	Yes
(C)	Yes	No	Yes	Yes	No
(D)	Yes	No	Yes	Yes	No
(E)	Yes	Yes	Yes	No	No

Answer: E. One of the most critical actions of gastrointestinal hormones is their trophic activity. Gastrin can stimulate mucosal growth throughout the gastrointestinal tract as well as growth of the exocrine pancreas. If most of the endogenous gastrin is removed by antrectomy, the gastrointestinal tract atrophies. Exogenous gastrin prevents the atrophy. Partial resection of the small intestine for tumor removal, morbid obesity, or other reasons results in hypertrophy of the remaining mucosa. The mechanism for this adaptive response is poorly understood. Both cholecystokinin and secretin stimulate growth of the exocrine pancreas. GLIP (glucose-dependent insulinotropic peptide) and motilin do not appear to have trophic actions on the gastrointestinal tract.

167. A 19-year-old man is fed intravenously for several weeks following a severe automobile accident. The intravenous feeding leads to atrophy of the gastrointestinal mucosa most likely because the blood *level* of which of the following hormones is reduced?

(A) Cholecystokinin only
(B) Gastrin only
(C) Secretin only
(D) Gastrin and cholecystokinin
(E) Gastrin and secretin
(F) Secretin and cholecystokinin

Answer: B. Gastrin has a critical role to stimulate mucosal growth throughout the gastrointestinal system.

Duodenum

168. Removal of proximal segments of the small intestine would most likely result in a decrease in which of the following?

(A) Basal acid output
(B) Maximal acid output
(C) Gastric emptying of liquids
(D) Gastric emptying of solids
(E) Pancreatic enzyme secretion

Answer: E. Inflammation or removal of the upper small intestine leads to a decrease in pancreatic and hepatobiliary function. The proximal small intestine contains a number of receptors that monitor the physical (volume) and chemical (pH, fat content, caloric density, osmolality) composition of the chyme emptied from the stomach. Stimulation of these receptors releases secretin, which acts on pancreatic ductal cells to increase HCO_3^- secretion, as well as cholecystokinin, which acts on pancreatic acinar cells to increase pancreatic enzyme secretion (lipases, amylases, and proteases). Stimulation of proximal small intestine receptors also activates neural reflexes, which initiate pancreatic enzyme and bicarbonate secretion, stimulate gallbladder emptying, and provide feedback for inhibitory regulation of gastric function. Removal of these reflexes decreases pancreatic secretion and gallbladder emptying and increases gastric emptying and acid output.

Carbonic Anhydrase

169. Carbonic anhydrase is active in high concentration in the:

(A) Chief cells of the stomach
(B) Parietal cells of the stomach
(C) Duodenal mucosa
(D) Pancreatic duct cells

Answer: B. Carbonic anhydrase is required for the conversion of CO_2 into H^+ and HCO_3^-, thus this enzyme will be present in the parietal cells in the stomach and the pancreatic cells.

Gastrointestinal Physiology

Gastrointestinal Digestion and Absorption

Optimum Environments for Digestion/Absorption

170. Enzymatic digestion in alkaline or neutral environment.

(A) Carbohydrates
(B) Proteins
(C) Carbohydrate and protein
(D) Triglycerides
(E) Carbohydrates, proteins, triglycerides

Answer: E. Digestion in the small intestine takes place at a pH which is close to neutral. Protein is also digested in the acid medium of the stomach.

Carbohydrate Digestion/Absorption

171. Which of the following is the main digestible carbohydrate normally consumed in the human diet?

(A) Amylose (B) Cellulose (C) Maltose (D) Starch

Answer: D. The three major sources of carbohydrates in the normal human diet include sucrose mainly from sugarcane and sugar beets, lactose from milk, and a wide variety of large polysaccharides collectively known as starches. Although some diets may contain a large quantity of cellulose, this substance cannot be digested by the human gut and is not considered a food. Maltose is a product of the digestion of starch but is not consumed in large quantities in the human diet.

172. Which of the following food substances requires chewing for digestion?

(A) Cheese (B) Eggs (C) Vegetables (D) Meat

Answer: C. Chewing is important for the digestion of all solid foods. It is especially important for most fruits and vegetables because these foods have an indigestible cellulose membrane surrounding the nutrient portion that must be broken to allow contact with digestive enzymes.

173. Which of the following substances must be further digested before it can be absorbed by specific carriers in intestinal cells?

(A) Fructose
(B) Sucrose
(C) Alanine
(D) Dipeptides
(E) Tripeptides

Answer: B. Only monosaccharides can be absorbed by intestinal epithelial cells. Disaccharides, such as sucrose, must be digested to monosaccharides before they are absorbed. On the other hand, proteins are hydrolyzed to amino acids, dipeptides, or tripeptides, and all three firms are transported into intestinal cells for absorption.

174. A patient deficient in sucrase has a primary abnormality that involves the:

(A) Parietal cells of the stomach
(B) Pancreas
(C) Small intestinal epithelium
(D) Salivary glands
(E) Liver

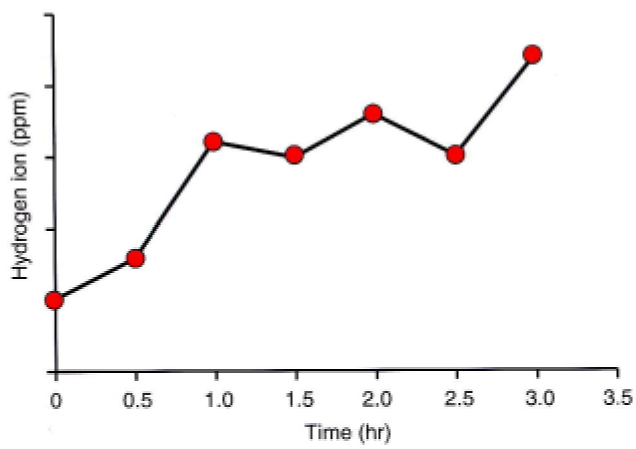
Answer: C. Sucrase is one of the required small intestinal brush border enzymes.

175. An 18-year-old college student reports that she experiences severe abdominal bloating and diarrhea within 1 hour of consuming dairy products. A subsequent hydrogen breath test is abnormal. The diarrhea and bloating can best be explained by which of the following?

(A) A deficiency in the brush border enzyme lactase
(B) Carbohydrate-induced secretory diarrhea
(C) Decreased intestinal surface area
(D) Decreased carbohydrate absorption
(E) A decrease in exocrine pancreatic secretion

Answer: A. Lactase is a brush border enzyme that hydrolyzes lactose into glucose and galactose. Patients with a lactase deficiency may experience diarrhea, cramps, and intestinal gas. The diarrhea and cramping reflect the osmotic effect of the sugar on water flux across the intestine. Colonic bacteria metabolize lactose to fatty acids, CO_2, and H_2.

176. A 19-year-old woman visits her physician because of nausea, diarrhea, light headedness, and flatulence. The physician administers 50 g of oral lactose at time zero, and measures breath hydrogen every 30 min for 3 hours using a hand-held monitor. The results are shown below. Which of the following best describes this patient's condition?



(A) Brush border lactase deficiency
(B) Brush border lactase excess
(C) Decreased pancreatic lactase
(D) Increased pancreatic lactase
(E) No digestive abnormality

Answer: A. Patients with a lactase deficiency cannot digest milk products containing lactose (milk sugar). The operons of gut bacteria quickly switch over to lactose metabolism, which results in fermentation that produces copious amounts of gas (a mixture of hydrogen, carbon dioxide, and methane). This gas, in turn, may cause a range of abdominal symptoms, including stomach cramps, bloating, and flatulence. The gas is absorbed by blood (especially in the colon), and exhaled from the lungs.

Celiac Disease

177. A 43-year-old woman presents with chief complaints of bulky and frequent diarrhea and weight loss. She experiences recurrent episodes of abdominal distension terminated by passage of stools. Laboratory data reveals a microcytic anemia, decreased serum calcium, and decreased serum albumin. After additional tests she is diagnosed with gluten-sensitive enteropathy. Her generalized decrease in intestinal absorption can be attributed to which of the following?

(A) Decreased intestinal motility
(B) Increased migrating motor complexes
(C) Decreased intestinal surface area
(D) Increased enterohepatic bile circulation
(E) Decreased gastric emptying

Answer: C. Gluten-sensitivity enteropathy, also known as celiac sprue, is characterized by an autoimmune-induced decrease in the absorptive surface area of the small intestine in response to gluten and other proteins in grain foods. In addition to a decrease in the area available for absorption of nutrients, minerals, electrolytes, and water, the membrane transporters of the remaining villous tip cells are impaired or absent.

178. A 45-year-old man adds lots of high-fiber wheat and bran foods to his diet to reduce his cholesterol. He loses 30 lb on the new diet, but has undesirable side effects such as stomach cramps, flatulence, and diarrhea. His gastroenterologist diagnoses a syndrome called gluten-enteropathy or celiac sprue. Which of the following is decreased in this man?

(A) Absorption of nutrients
(B) Digestion of fat
(C) Stool carbohydrates
(D) Stool fat
(E) Stool nitrogen

Answer: A. Celiac sprue is a chronic disease of the digestive tract that interferes with the absorption of nutrients from food. The mucosal lesions seen on upper GI biopsy are the result of an abnormal, genetically determined, cell-mediated immune response to gliadin, a constituent of the gluten round in wheat. A similar response occurs to comparable proteins found in rye and barley. Gluten is not found in oats, rice, and maize. When affected individuals ingest gluten, the mucosa of their small intestine is damaged by an immunologically mediated inflammatory response, resulting in malabsorption and maldigestion at the brush border. Digestion of fat is normal in celiac sprue because the pancreas which secretes lipase still functions normally. Because celiac sprue causes malabsorption it should be clear that the stool content of carbohydrates, fat, and nitrogen is increased.

179. A 10-year-old boy presents with below-average body weight and height, signs of vitamin K deficiency, steatorrhea, and bloating. He is found to have the MHC class II antigen HLA-DQ2. Which of the following is the most appropriate dietary treatment for malabsorption in this condition?

(A) Fat-free diet
(B) Lactose-free diet
(C) Gluten-free diet
(D) High-fiber diet
(E) Low-salt diet

Answer: C. Malabsorption syndrome refers to the inability to adequately absorb nutrients and vitamins from the intestinal tract. One example of a malabsorption syndrome is the autoimmune disease, celiac sprue, which is also called gluten enteropathy. The disease is characterized by a deficiency in MHC class II antigen HLA-DQ2, which causes an allergic response to ingestion of gluten and related proteins. Elimination of these proteins, which are found in wheat, rye, barley, and oats, can restore normal bowel function in these patients.

Protein Digestion/Absorption

180. Which of the following statements most accurately describes amino acid uptake from the intestinal lumen? Amino acid:

(A) Crosses the mucosal membrane by a process of passive diffusion
(B) Crosses the mucosal membrane by a carrier-mediated transport process requiring the concomitant transfer of Na^+ in the same direction
(C) Crosses the serosal membrane by a carrier-mediated transport process requiring the concomitant transfer of Na^+ in the opposite direction
(D) Is actively transported across the serosal membrane against a concentration gradient
(E) Uptake is independent of high-energy phosphate bonds in the cell

Answer: B. Amino acids are generally initially absorbed across the luminal membrane via secondary active transport linked to sodium and across the serosal membrane by simple/facilitated diffusion, although other processes are probably involved as well.

181. Digestion of which of the following foodstuffs is impaired to the greatest extent in patients with achlorhydria?

(A) Carbohydrate
(B) Fat
(C) Protein

Answer: C. Achlorhydria means simply that the stomach fails to secrete hydrochloric acid; it is diagnosed when the pH of the gastric secretions fails to decrease below 4 after stimulation by pentagastrin. When acid is not secreted, pepsin also usually is not secreted; even when it is, the lack of acid prevents it from functioning because pepsin requires an acid medium for activity. Thus, protein digestion is impaired.

Carrier-Transport

182. Absorbed by secondary active transport.

(A) Glucose
(B) Amino acids
(C) Glucose and amino acids
(D) Triglycerides
(E) Glucose, amino acids, and triglycerides

Answer: C. Both glucose and amino acids are absorbed across the luminal membrane via secondary active transport linked to sodium.

183. Absorption of digestive products involves carrier-mediated transport in small intestine.

(A) Carbohydrates
(B) Proteins
(C) Carbohydrate and protein
(D) Triglycerides
(E) Carbohydrates, proteins, triglycerides

Answer: C. The end products of carbohydrate and protein digestion are absorbed across the luminal membrane via secondary active transport. CHO absorption also involves facilitative transport.

184. Absorption by small intestine utilizes Na^+ carrier.

(A) Glucose
(B) Amino acids
(C) Glucose and amino acids
(D) Triglycerides
(E) Glucose, amino acids, and triglycerides

Answer: C. Both glucose and amino acids are absorbed from the small intestine via secondary active transport linked to sodium. An increase in intraluminal sodium will accelerate the uptake of glucose and amino acids. The reverse is also true; low intraluminal sodium would slow the absorption of glucose and amino acids.

185. Which of the following is transported in intestinal epithelial cells by a Na^+-dependent cotransport process?

(A) Fatty acids
(B) Triglycerides
(C) Fructose
(D) Alanine
(E) Oligopeptides

Answer: D. Fructose is the only monosaccharide that is not absorbed by Na^+-dependent cotransport; it is transported by facilitated diffusion. Amino acids are absorbed by Na^+-dependent cotransport, but oligopeptides (larger peptide units) are not. Triglycerides are not absorbed without further digestion. The products of lipid digestion, such as fatty acids, are absorbed by simple diffusion.

186. A Na^+-dependent active transport system is necessary for the intestinal absorption of all of the following EXCEPT…

(A) Dipeptides
(B) Bile salts
(C) Fructose
(D) Vitamin C
(E) Glucose

Answer: A and C. Unlike glucose, which depends on a Na^+-dependent active transport system for absorption, fructose is absorbed by facilitated diffusion. Active transport for fructose is not required because the intracellular fructose concentration is maintained at a low value by intracellular enzymes that rapidly convert fructose to glucose and lactose. Bile salts, and vitamin C all rely on Na^+-dependent active transport systems to be absorbed. Dipeptides are absorbed with H^+.

187. Na^+-dependent transport is responsible for the absorption of all of the following EXCEPT…

(A) Vitamin E
(B) Amino acids
(C) Glucose
(D) Bile salts

Answer: A. Amino acids and glucose are absorbed in the proximal intestine by Na^+-dependent active transport systems. Bile salts are reabsorbed by a Na^+-dependent active transport system located within the terminal ileum. Vitamin E is a fat-soluble vitamin and is absorbed by passive diffusion along with other lipids in the proximal intestine.

188. Moved from intestinal epithelial cell to intracellular space by diffusion.

(A) Glucose
(B) Amino acids
(C) Glucose and amino acids
(D) Triglycerides
(E) Glucose, amino acids, and triglycerides

Answer: C. Both glucose and amino acid are transported out to the intracellular space by facilitative diffusion but a small amount of simple diffusion does occur. Triglycerides form chylomicrons and transported out by exocytosis.

189. Absorption used same carrier as galactose.

(A) Glucose
(B) Amino acids
(C) Glucose and amino acids
(D) Triglycerides
(E) Glucose, amino acids, and triglycerides

Answer: A. There are not separate carriers for glucose and galactose. They must compete for the same carrier. Adding galactose will slow the transport of glucose and vice versa.

190. Rate of absorption of glucose from the small intestine can be reduced by presence in the small intestine of:

(A) Mannose
(B) Fructose
(C) Xylose
(D) Galactose
(E) Arabinose

Answer: D. Since glucose and galactose compete for the same carrier, the presence of galactose will decrease the absorption rate of glucose. Fructose is absorbed by a mechanism completely independent from glucose and galactose. A, C, and E are not monosaccharides.

191. The addition of glucose to the lumen of an intestinal segment is found to increase the potential difference across the mucosa. The addition of compound X to the lumen decreases the absorption of glucose but causes a further increase in the potential difference. Compound X is most likely:

(A) Alanine
(B) Fructose
(C) Galactose
(D) Palmitic acid
(E) Glycerol

Answer: C. Glucose and galactose compete for the entry of SGLT-1, which is coupled with Na^+ absorption. Thus, adding galactose increased the “sugar” concentration and SGLT-1 absorbed more Na^+, but decreased glucose absorption with expense of increased galactose absorption.

192. A newborn with severe diarrhea is found to have an inherited defect in a glucose transporter resulting in glucose/galactose malabsorption, necessitating a glucose- and galactose-free diet. Which of the following is the transport protein responsible for entry of glucose into the intestinal enterocyte?

(A) GLUT-2
(B) GLUT-5
(C) SGLT 1
(D) SGLT 2
(E) SGLT 5

Answer: C. The transport protein responsible for the sodium-dependent glucose transport in the small intestine is termed the SGLT1 (Na^+-glucose transporter). The absorption of glucose occurs through the coordinated action of transport proteins located in the brush border and basolateral membranes of the enterocyte. Glucose uptake into the enterocyte from the lumen of the GI tract occurs primarily via the sodium-dependent SGLT1 secondary active transport mechanism. Glucose exit from the enterocyte into the extracellular fluid occurs by facilitated diffusion and is mediated by the membrane transporter, GLUT-2. The Na^+-glucose cotransporter also transports galactose. GLUT-5 is the membrane transporter located on the apical portion of the enterocyte responsible for the facilitated entry of fructose into the cell.

Thus, when the cotransporter is congenitally defective, the resulting glucose and galactose malabsorption causes severe diarrhea that can be fatal if glucose and galactose are not removed from the diet. A similar secondary active transport process (Na^+-glucose cotransport) occurs in the renal tubules via SGLT1 and SGLT2.

Fat Digestion/Absorption

193. Bile salts promote absorption of lipids as a result of their ability to:

(A) Form micelles or water soluble complexes
(B) Increase secretion of pancreatic lipase
(C) Activate pancreatic lipase
(D) Stimulate gallbladder contraction
(E) Increase surface tension of fat particles

Answer: A.

194. Absorption only after formation of soluble complexes with bile salts.

(A) Carbohydrates
(B) Proteins
(C) Carbohydrate and protein
(D) Triglycerides
(E) Carbohydrates, proteins, triglycerides

Answer: D. Bile salts are important for the digestion and the absorption of lipid-soluble substances, not the water-soluble products of carbohydrate and protein digestion.

195. Which of the following is required for absorption of the fat soluble vitamins contained in his supplement?

(A) Intrinsic factor
(B) Chymotrypsin
(C) Pancreatic lipase
(D) Pancreatic amylase
(E) Secretin

Answer: C. Absorption of the fat-soluble vitamins (A, D, E, and K) is diminished if there is a lack of pancreatic lipase. Lipase is required to produce monoglycerides that, in combination with bile salts, make it possible to bring the fat-soluble vitamins close to the mucosal cell surface for absorption. With the exception of vitamin B_{12}, which is absorbed bound to intrinsic factor in the ileum, vitamins are absorbed chiefly in the upper small intestine.

196. The major lipolytic digestive enzymes in man:

(A) Are found in gastric juices
(B) Are activated by enterokinase
(C) Are potentiated by bile salts
(D) Produce 2-monoglycerides, free fatty acids, and glycerol
(E) Require a pH of greater than 10 for optimal activity

Answer: C and D. Most lipases are secreted by the pancreas as active enzymes. The major end products are 2-monoglycerides and free fatty acids. However, some triglycerides are completely digested to fatty acids and glycerol. Enterokinase is responsible for the activation of trypsinogen to trypsin. All of this takes place in the fairly neutral pH of the small intestine.

197. Micelle formation is necessary for the intestinal absorption of

(A) Glycerol
(B) Galactose
(C) Leucine
(D) Bile acids
(E) Vitamin B12
(F) Vitamin D

Answer: F. Micelles provide a mechanism for solubilizing fat soluble nutrients in the aqueous solution of the intestinal lumen until the nutrients can be brought into contact with and absorbed by the intestinal epithelial cells. Because vitamin D is fat soluble, it is absorbed in the same way as other dietary lipids. Glycerol is one product of lipid digestion that is water soluble and is not included in micelles. Galactose and leucine are absorbed by Na^+-dependent cotransport. Although bile acids are a key ingredient of micelles, they are absorbed by a specific Na^+-dependent cotransporter in the ileum. Vitamin B_{12} is water soluble; thus, its absorption does not require micelles.

198. Micelle formation is necessary for absorption of…

(A) Bile salts
(B) Iron
(C) Cholesterol
(D) Alcohol
(E) B vitamins

Answer: C. Micelles are necessary for absorption of dietary lipids such as cholesterol. Bile salts, iron, and, B vitamins are absorbed by membrane-bound active transport systems. Alcohol is both water- and fat-soluble and so can diffuse directly across the membranes.

199. Micelle formation is necessary for absorption and transport to intestinal epithelium of:

(A) Cholesterol
(B) Short-chain fatty acids
(C) Vitamin D
(D) Glycerol

Answer: A and C. The correct answers would include any fat-soluble substances.

200. Substances that enter the circulation as chylomicrons via lymphatic ducts are:

(A) Cholesterol
(B) Short-chain fatty acids
(C) Vitamin K
(D) Glycerol

Answer: A and C. These would include lipid-soluble but not water-soluble compounds. Cholesterol and vitamin K are lipid-soluble compounds. Short-chain fatty acids are fairly water soluble and can diffuse directly into the systemic circulation. Long-chain fatty acids, on the other hand, being lipid soluble, must enter as part of the chylomicrons via the lymphatics. Glycerol is a water-soluble compound.

201. Dietary fat, after being processed, is extruded from the mucosal cells of the gastrointestinal tract into the lymphatic ducts in the form of which of the following?

(A) Monoglycerides
(B) Diglycerides
(C) Triglycerides
(D) Chylomicrons
(E) Free fatty acids

Answer: D. Long-chain fatty acids are extruded from enterocytes in the form of chylomicrons into the lymphatic system. Triglycerides are hydrolyzed to monoglycerides and taken into mucosal cells. lf the fatty acids are short chains (less than 10-12 carbon atoms), they are extruded in the form of free fatty acids into the portal blood. Chylomicrons represent triglycerides and esters of cholesterol that have been invested in the intestinal mucosa with a coating of phospholipid, protein, and cholesterol.

202. Mainly transported to blood vessel by lacteal (lymphatic capillaries).

(A) Glucose
(B) Amino acids
(C) Glucose and amino acids
(D) Triglycerides
(E) Glucose, amino acids, and triglycerides

Answer: D. A major point was that the chylomicrons containing the end products of lipid absorption are absorbed into the lymphatic and not the general circulation. The products of carbohydrate and protein digestion are absorbed directly into the systemic capillaries.

203. Fats are transported from intestinal cells to blood plasma primarily in the form of…

(A) Micelles
(B) Chylomicrons
(C) Triglycerides
(D) Fatty acids
(E) Monoglycerides

Answer: B. Chylomicrons are small lipid droplets within the enterocytes that are reconstituted from the lipid digestion products formed in the intestine. The chylomicrons are transported from the enterocytes by exocytosis by the intestinal fluid surrounding the intestine. The chylomicrons are then absorbed into the intestinal lacteals, from which they enter the circulation.

204. Long-chain fatty acids absorbed by the intestinal mucosa cells reach the blood in the form of:

(A) Monoglycerides
(B) Diglycerides
(C) Micelles
(D) Chylomicrons
(E) Free fatty acids

Answer: D. Absorbed long-chain fatty acids are resynthesized to triglycerides, which then form a major part of the chylomicron.

205. Materials whose intestinal absorption is greatly aided by the presence of bile salt and

phospholipids:

(A) Cholesterol
(B) Vitamin B_{12}
(C) Long-chain fatty acids
(D) Short-chain fatty acids

Answer: A and C. Correct answers here would include any lipid substances. Vitamin B_{12} is absorbed via intrinsic factor, and not lipid soluble. Short-chain fatty acids are water soluble.

206. Which of the following is not normally found in abundance in the portal blood?

(A) Amino acids
(B) Glucose
(C) Short-chain fatty acids
(D) Triglycerides

Answer: D. Triglycerides are digested within the intestinal lumen to monoglycerides and free fatty acids, which are then absorbed directly through the membrane of the intestinal epithelial cells. After entering the epithelial cell, the fatty acids and monoglycerides are taken up by the cell's smooth endoplasmic reticulum where they are mainly used to form new triglycerides that are subsequently released in the form of chylomicrons through the base of the epithelial cell. The chylomicrons are absorbed by the central lacteal in the villus and transported in lymph through the thoracic lymph duct to the circulating blood. Thus, most triglycerides bypass the portal circulation.

Fat Malabsorption

207. A 26-year-old man presents to the emergency room with a 48-hour bout of diarrhea with steatorrhea. Which of the following best accounts for the appearance of excess fat in the stool?

(A) Decreased bile salt pool size
(B) Delayed gastric emptying
(C) Decreased gastric acid secretion
(D) Decreased secretion of intrinsic factor
(E) Decreased gastric accommodation

Answer: A. Steatorrhea is defined as excess loss of fat in the stool. Numerous pathophysiological situations can cause the loss of excess fat in the stool including a decrease in bile acid pool size, inactivation or decreased intraluminal concentration of pancreatic lipase in the small intestine, and decreased intestinal absorptive surface area. A decrease in bile acid pool size results in an increased delivery of fats into the colon, which in turn inhibits fat absorption and promotes water secretion.

208. A 32-year-old woman presents to the emergency department with abdominal pain and diarrhea accompanied by steatorrhea. Gastric analysis reveals a basal acid output (I2 mmol/h) greater than normal (< 5 mmol/h). The steatorrhea is most likely due to which of the following?

(A) Inactivation of pancreatic lipase
(B) Delayed gastric emptying
(C) Decreased gastric acid secretion
(D) Decreased secretion of intrinsic factor
(E) Decreased pyloric sphincter tone

Answer: A. The process of fat digestion begins in the stomach and is completed in the proximal small intestine, predominately by enzymes synthesized and secreted by the pancreatic acinar cells. The major lipolytic pancreatic enzyme is the carboxylic esterase, known as lipase. Full activity requires the protein cofactor colipase, bile salts, fatty acids, as well as an alkaline pH. Excess delivery of acid into the proximal small intestine leads to reduced lipolytic activity.

209. A 49-year-old male patient with severe Crohn's disease has been unresponsive to drug therapy and undergoes ileal resection. After the surgery, he will have steatorrhea because

(A) The liver bile acid pool increases
(B) Chylomicrons do not form in the intestinal lumen
(C) Micelles do not form in the intestinal lumen
(D) Dietary triglycerides cannot be digested
(E) The pancreas does not secrete lipase

Answer: C. Ileal resection removes the portion of the small intestine that normally transports bile acids from the lumen of the gut and recirculates them to the liver. Because this process maintains the bile acid pool, new synthesis of bile acids is needed only to replace those bile acids that are lost in the feces. With ileal resection, most of the bile acids secreted are excreted in the feces, and the liver pool is significantly diminished. Bile acids are needed for micelle formation in the intestinal lumen to solubilize the products of lipid digestion so that they can be absorbed (TGs still can be digested but not properly). Chylomicrons are formed within the intestinal epithelial cells and are transported to lymph vessels.

210. A 67-year-old man with a history of alcohol abuse presents to the emergency room with severe epigastric pain, hypotension, abdominal distension, and diarrhea with steatorrhea. Serum amylase and lipase are found to be greater than normal, leading to a diagnosis of pancreatitis. The steatorrhea can be accounted for by a decrease in the intraluminal concentration of which of the following pancreatic enzymes?

(A) Amylase
(B) Trypsin
(C) Chymotrypsin
(D) Lipase
(E) Colipase

Answer: D. The process of fat digestion begins in the stomach and is completed in the proximal small intestine, predominately by enzymes synthesized and secreted by the pancreatic acinar cells. The major lipolytic pancreatic enzyme is the carboxylic esterase, known as lipase. Full activity requires the protein cofactor colipase, as well as an alkaline pH, bile salts, and fatty acids.

Duodenum

211. Digestion is partially completed before entering duodenum.

(A) Carbohydrates
(B) Proteins
(C) Carbohydrate and protein
(D) Triglycerides
(E) Carbohydrates, proteins, triglycerides

Answer: C. The digestion of CHO starts in the mouth by salivary α-amylase. Protein digestion begins in the stomach by pepsin secreted from gastric glands.

212. With respect to cobalamin-intrinsic factor binding in a normal individual, nearly all binding of cobalamin to intrinsic factor occurs in which of the following organs?

(A) Stomach
(B) Duodenum
(C) Jejunum
(D) Ileum
(E) Colon

Answer: B. Although intrinsic factor is secreted by the parietal cells of the gastric mucosa, binding to cobalamin occurs in the duodenum. Cobalamin, also known as vitamin B_{12}, is provided almost entirely from animal products in the human diet. Gastric digestion of food liberates cobalamin where, at low pH, it binds primarily to R-binder protein, derived primarily from salivary secretions. In the duodenum, pancreatic proteases release cobalamin from the R-binder protein where cobalamin then rapidly complexes with intrinsic factor and is transported along the gut to the terminal ileum, where specific receptors located on the villus tip cells bind the cobalamin-intrinsic factor complex.

Iron Absorption

213. Where does the absorption of iron occur?

(A) Fundus of the stomach
(B) Antrum of the stomach
(C) Duodenum of the intestine
(D) Ileum of the intestine
(E) Colon

Answer: C. Iron is absorbed primarily within the duodenum and jejunum.

214. A 47-year-old woman with hypermenorrhea develops an iron-deficiency anemia requiring iron supplements. Which of the following best describes iron digestion and absorption?

(A) About 100 mg of iron is absorbed per day.
(B) Iron is absorbed rapidly from the small intestine.
(C) Iron is transported in the blood bound to transferrin.
(D) In general, iron must be oxidized from the ferrous to the ferric state for efficient absorption.
(E) Iron is transported into enterocytes by a ferroportin transporter on the apical membrane.

Answer: C. Iron is transported in the blood bound to the β globulin, transferrin. Excess iron is stored in all cells, but especially in hepatocytes where it combines with apoferritin. The stored form is called ferritin. The rate of iron absorption is extremely slow, with a maximum of only a few milligrams per day. Iron is absorbed primarily in the ferrous form. Therefore, ferrous iron compounds, rather than ferric compounds, are effective in treating iron deficiency. Ferroportin is located on basolateral membrane.

215. A 23-year-old woman complains of abdominal cramps and bloating that are relieved by defecation. Subsequent clinical evaluation reveals an increased maximal acid output, decreased serum calcium and iron concentrations, and microcytic anemia. Inflammation in which area of the GI tract best explains these findings?

(A) Stomach
(B) Duodenum
(C) Jejunum
(D) Ileum
(E) Colon

Answer: B. Inflammation of the duodenum may lead to increased acid output, hypocalcemia, and microcytic anemia. Increased basal and maximal acid outputs may result from excessive stimulation of the parietal cell (e.g., hypergastrinemia) or reduced inhibitory feedback (i.e., reduced effect of enterogastrone and the enterogastric reflex). The latter may occur when the proximal small intestine is inflamed. Although calcium is absorbed along the entire length of the small intestine, it is absorbed primarily in the duodenum. Similarly, iron is absorbed primarily in the duodenum and the microcytic anemia is the result of reduced stores of iron.

Calcium Absorption

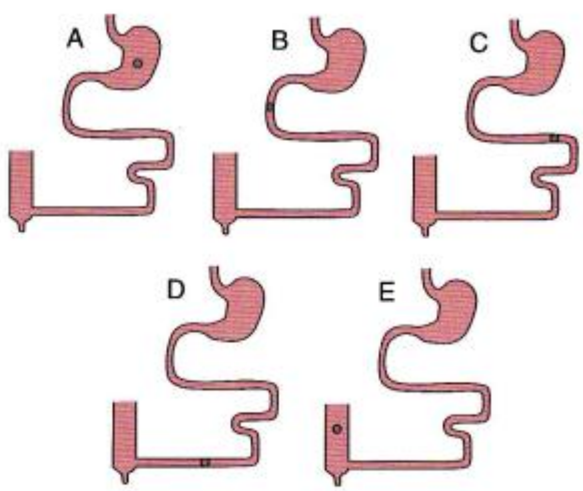
216. Calcium absorption in the small intestine is increased by:

(A) Vitamin D
(B) Parathyroid hormone
(C) Hypocalcemia
(D) All of the above
(E) None of the above

Answer: A. A major action of vitamin D is the absorption of calcium and phosphate from the small intestine. The small intestine is not a target tissue for parathyroid hormone. In hypocalcemia there can be increased absorption of calcium from the small intestine, but this will occur only if the activity of vitamin D increases.

Ileum

217. A 48-year-old man consumes a healthy meal. At which of the following locations is vitamin B_{12} most likely to be absorbed?



Answer: D. Vitamin B_{12} is absorbed primarily by the ileum.

218. Major site of vitamin B_{12} absorption is:

(A) Stomach
(B) Duodenum
(C) Jejunum
(D) Ileum
(E) Descending colon

Answer: D. Vitamin B_{12} attached to intrinsic factor is absorbed in the distal ileum. A nonfunctional distal ileum will lead to a vitamin B_{12} deficiency and anemia.

219. An 18-year-old man with pernicious anemia lacks intrinsic factor, which is necessary for the absorption of cyanocobalamin. Vitamin B_{12} is absorbed primarily in which portion of the GI tract?

(A) Stomach
(B) Duodenum
(C) Jejunum
(D) Ileum
(E) Colon

Answer: D. Most vitamins are absorbed in the upper small intestine, but vitamin B_{12} (cobalamin) is absorbed primarily in the terminal ileum. Vitamin B_{12} binds with intrinsic factor a glycoprotein secreted by the parietal cells of the gastric mucosa. The vitamin B_{12}-intrinsic factor complex is propelled along the small intestine to the terminal ileum, where specific active transporters located on the enterocyte microvilli bind the vitamin B_{12}-intrinsic factor complex and the complex is absorbed across the ileal mucosa. Pernicious anemia is a disease in which there is autoimmune destruction of the parietal cells. Vitamin B_{12} can also be produced by gastrectomy with removal of the intrinsic factor-secreting tissue or by diseases of the terminal ileum. Binding of the Vitamin B_{12}-intrinsic factor complex requires Ca^{2+}. Whereas vitamin B_{12} and folate absorption are Na^+ independent, all seven of the other water-soluble vitamins are absorbed by carriers that are Na^+ cotransporters.

220. A 48 year old female presents to the physician with severe diarrhea and cramping. She has had these complaints for last five years. She is diagnosed with inflammatory bowel disease and undergoes resection of terminal ileum. She is subsequently likely to develop a deficiency of which of the following?

(A) Iron
(B) Calcium
(C) Vitamin B_{12}
(D) Vitamin D
(E) Vitamin A

Answer: C. Vitamin B_{12} is absorbed in ileum. Iron and calcium are absorbed in duodenum. Vitamin A and D are fat soluble and thus absorbed with fats along the small intestine.

221. Which of the following is a likely consequence of ileal resection?

(A) Achalasia
(B) Atrophic gastritis
(C) Constipation
(D) Peptic ulcer
(E) Vitamin B_{12} deficiency

Answer: E. Vitamin B_{12} absorption requires intrinsic factor, which is a glycoprotein secreted by parietal cells in the stomach. Binding of intrinsic factor to dietary vitamin B_{12} is necessary for attachment to specific receptors located in the brush border of the ileum. Therefore, ileal resection can lead to vitamin B_{12} deficiency. Achalasia is a neuromuscular failure of relaxation at the lower end of the esophagus with progressive dilation, tortuosity, incoordination of peristalsis, and often hypertrophy of the proximal esophagus. Atrophic gastritis is a type of autoimmune gastritis that is mainly confined to the acid-secreting corpus mucosa. The gastritis is diffuse and eventually severe atrophy develops. Ileal resection is likely to cause diarrhea, but not constipation. Benign gastric and duodenal ulcers are best classified together as peptic ulcers even though their etiology is different. In both types of ulcer it is acid and pepsin which causes the mucosal damage. Duodenal ulcers are more common.

222. After removal of the ileum, one observes a decrease:

(A) In size of bile acid pool
(B) In fat content of feces
(C) In absorption of vitamin B_{12}
(D) In fecal volume

Answer: A and C. To be more specific, if the distal ileum is removed, intrinsic factor with attached vitamin B_{12} cannot be absorbed and bile salts cannot be reabsorbed and recycled. Since the liver has a limited capacity to synthesize replacements, there will usually be a bile salt deficiency. This will disrupt the digestion and absorption of most lipid substances and increase the fat content of the stool.

223. Ileal resection:

(A) Decreases the proportion of primary bile acids in bile
(B) Increases the synthesis of bile acids by the liver
(C) Increases the rate of biliary fluid secretion
(D) Will not affect dietary lipid absorption
(E) Decreases fecal bile acid excretion

Answer: B. Ileal resection compromises the reabsorption (increases fecal loss) and recycling of bile, thus liver has to increase the synthesis of bile acid and increases the proportion of primary bile acids in bile. Majority of bile are recycled bile and thus, ileal resection decreases the rate of biliary fluid secretion and thus decreases dietary lipid absorption.

224. A 37-year-old man is admitted to the hospital due to an exacerbation of his Crohn disease with severe inflammation of the ileum. Which of the following would be seen?

(A) Increased vitamin B_{12} absorption
(B) Decreased bile acid pool size
(C) Increased colon absorption of water
(D) Decreased release of secretin
(E) Increased absorption of dietary fats

Answer: B. Individuals with inflammatory disease of the ileum have decreased bile acid pool size due to decreased bile acid reabsorption.

This results in reduced absorption of dietary triglycerides and fat-soluble vitamins, including vitamin K. The efficient absorption of dietary fats requires the presence of critical concentrations of primary and secondary bile salts (1-5 mmol/L). Because the absolute amount of bile acids available for fat digestion during a meal (the bile acid pool size) is generally less than the amounts required for complete digestion and absorption, bile acids must be recirculated via the enterohepatic circulation. Conservation of bile acids during a meal is highly efficient and occurs primarily from the distal ileum via sodium-dependent, secondary active transport. The increased delivery of dietary fat and bile acids into the colon decreases colonic absorption of water. The loss of bile salts in the stool cannot be fully compensated for by increased hepatic synthesis, and, thus, there is a resultant decrease in bile acid pool size.

225. A 57-year-old man undergoes resection of the distal 100 cm of the terminal ileum as part of treatment for Crohn disease. The patient likely will develop malabsorption of which of the following?

(A) Iron
(B) Folate
(C) Lactose
(D) Bile salts
(E) Protein

Answer: D. The terminal ileum contains specialized cells responsible for the absorption of primary and secondary bile salts by active transport. Bile salts are necessary for adequate digestion and absorption of fat. Resection of the ileum prevents the absorption of bile acids, which leads to steatorrhea. In addition, diarrhea results because the unabsorbed bile acids enter the colon where they increase adenylate cyclase activity, thus promoting the secretion of water into the lumen of the colon causing an increase in the water content of the feces. Iron, folate, carbohydrate, and protein absorption occur primarily in the upper portions of the small intestine.

226. A 57-year-old woman undergoes resection of the terminal ileum as part of treatment for her chronic inflammatory bowel disease. Removal of the terminal ileum will most likely result in which of the following?

(A) Increased bile acid concentration in the enterohepatic circulation
(B) Decreased glucose absorption
(C) Increased water content of the feces
(D) Increased iron absorption
(E) Increased fat absorption

Answer: C. Removal of the terminal ileum can lead to diarrhea and steatorrhea. The terminal ileum contains specialized cells responsible for the absorption of primary and secondary bile salts by active transport. Bile salts are necessary for adequate digestion and absorption of fat. In the absence of the terminal ileum there will be an increase in the amounts of bile acids and fatty acids delivered to the colon. Fats and bile salts in the colon increase the water content of the feces by promoting the influx (secretion) of water into the lumen of the colon. Glucose is absorbed in the duodenum/jejunum. Iron is primarily absorbed in the duodenum.

227. A 63-year-old woman presents with diarrhea, abdominal pain, and flushing. The urinary excretion of the serotonin metabolite, 5-hydroxyrndoleacetic acid (5-HIAA) is elevated. Abdominal CT reveals a tumor in the terminal ileum. Surgical resection of the terminal ileum will most likely result in which of the following?

(A) A decrease in absorption of amino acids
(B) An increase in the water content of the feces
(C) An increase in the concentration of bile acid in the enterohepatic circulation
(D) A decrease in the fat content of the feces
(E) An increase in the absorption of iron

Answer: B. Removal of the terminal ileum can lead to diarrhea and steatorrhea. The terminal ileum contains specialized cells responsible for the absorption of primary and secondary bile salts by active transport. Bile salts are necessary, for adequate digestion and absorption of fat. In the absence of the terminal ileum there will be an increase in the amounts of bile acids and fatty acids delivered to the colon. Fats and bile salts in the colon increase the ware content of the feces by promoting the influx (secretion) of water into the lumen of the colon. Amino acids are absorbed in the jejunum. Iron is primarily absorbed in the duodenum.

Gastrointestinal neuroendocrine tumors are derived from the diffuse neuroendocrine system of the GI tract, which is composed of amine- and acid-producing cells with different hormonal profiles, depending on the site of origin. The tumors they produce are generally divided into carcinoid tumors (ectodermal stem cells) and pancreatic endocrine tumors. One third of all primary gut tumors are carcinoid. Carcinoid tumors are frequently classified according to their anatomic area of origin (foregut, midgut, and hindgut). Small intestinal (midgut) carcinoid tumors arise from the argentaffin cells of the crypts of Lieberkühn the terminal ileum, and have high serotonin content. Small intestinal carcinoids are the most common cause of the carcinoid syndrome (cutaneous flushing, diarrhea, bronchospasm, right heart valvular lesions), which is manifest when they metastasize, but only occurs in 5% to 10% of carcinoid tumors.

Colon and Fluid Absorption

228. The normal volume of daily oral fluid intake:

(A) Typically appears in the feces each day
(B) Will cause severe diarrhea if it reaches the colon
(C) Is typically reabsorbed in the stomach
(D) Results in dumping syndrome if it reaches the jejunum
(E) Comprises less than half of the fluid absorbed per day

Answer: E. Normal diet usually contains 2 L of fluid. Feces usually contain 100 mL, and colonic reabsorption is somewhat close to the dietary intake (1.9 L), thus it won't cause diarrhea in healthy subject. Main function of stomach is secretion and reabsorption is close to nil. Dumping syndrome is the result of abnormally fast gastric emptying, and again, 2 L of fluid can be handled by colons.

229. The basic mechanism for fluid movement out of lumen of the intestine is thought to depend on active:

(A) Water transport and passive ion movements
(B) Cl^- transport, with Na+ and water following passively
(C) Na^+ transport by intestinal epithelium with water following passively
(D) HCO_3^- transport with water following passively
(E) None of the above.

Answer: C. The absorption of water and most electrolytes from the intestine is driven by the active reabsorption of sodium.

230. A patient has vomiting and severe watery diarrhea after eating spoiled shellfish. Intravenous fluid and electrolyte replacement was started, and a stool specimen was taken, which came back positive for *Vibrio cholerae*. Which of the following statements best describes water and electrolyte absorption in the GI tract?

(A) Osmotic equilibration of chyme occurs in the ileum.
(B) The small intestine and colon have similar absorptive capacities.
(C) Osmotic equilibration of chyme occurs in the stomach.
(D) The majority of absorption occurs in the jejunum.
(E) The incubation period for *V. cholerae* is 1 week.

Answer: D. Most water and electrolyte absorption occurs in the jejunum, with the duodenum serving primarily as the site of osmotic equilibration of chyme. Water absorption is passive and occurs as the direct result of active sodium absorption. The small intestine and colon absorb approximately 9 to 12 L of fluid per 24-hour period, most of which comes from gastrointestinal secretions. In contrast to the small intestine, the colon has a limited capacity to absorb water (approximately 3 to 6 L/d); most water absorption in the colon occurs in the proximal colon. The incubation period for *V. cholerae* is 24 to 48 hours.

231. The colon:

(A) Actively absorbs H_2O
(B) Actively absorbs Na^+ and Cl^-
(C) Has mass movements for mixing of contents
(D) Secretes HCO_3^- and K^+

Answer: B and D. The colon actively absorbs sodium and chloride and passively absorbs water. No cell actively absorbs water. The colon also demonstrates a net secretion of potassium and bicarbonate. This is important because with diarrhea, the loss of potassium can cause hypokalemia and the loss of bicarbonate can cause a metabolic acidosis. Mass movements are propulsive movements.

232. Which of the following best describes colonic function?

(A) Absorption of Na^+ in the colon is under hormonal (aldosterone) control.
(B) Bile acids enhance absorption of water from the colon.
(C) Net absorption of HCO_3^- occurs in the colon.
(D) Net absorption of K^+ occurs in the colon.
(E) The luminal potential in the colon is positive.

Answer: A. Both the absorption of Na^+ and secretion of K^+ from the colon are affected by changes in circulating levels of aldosterone. The major route of absorption of sodium in the colon is electrogenic transport. Because of the "tight" nature of the tight junctions that connect cells in the colon, a relatively large potential difference exists between the mucosal (negative) and serosal (positive) surfaces of the absorptive cells. This electrical difference favors the net secretion of K^+ into the lumen. Secretion of HCO_3^- occurs in exchange for absorption of Cl^-. No counterbalancing cation exchange pumps are present in the colon.

233. An 83-year-old woman with constipation is prescribed a high-fiber diet, which leads to an increased production of short-chain fatty acids (SCFAs). SCFA absorption occurs almost exclusively from which of the following segments of the GI tract?

(A) Stomach
(B) Duodenum
(C) Jejunum
(D) Ileum
(E) Colon

Answer: E. The colon is the major site for the generation and absorption of short-chain fatty acids. They are products of bacterial metabolism of undigested complex carbohydrates derived from fruits and vegetables. In addition to exhibiting trophic effects on the colonic mucosa, they are believed to promote sodium absorption from the colon. The mechanism of action remains controversial.

234. *Vibrio cholerae* causes diarrhea because it

(A) Increases HCO_3^- secretory channels in intestinal epithelial cells
(B) Increases Cl^- secretory channels in crypt cells
(C) Prevents the absorption of glucose and causes water to be retained in the intestinal lumen isosmotically
(D) Inhibits cyclic adenosine monophosphate (cAMP) production in intestinal epithelial cells
(E) Inhibits inositol 1,4,5-triphosphate (IP_3) production in intestinal epithelial cells

Answer: B. Cholera toxin activates adenylate cyclase and increases cyclic adenosine monophosphate (cAMP) in the intestinal crypt cells. In the crypt cells, cAMP activates the Cl^--secretory channels and produces a primary secretion of Cl^-, with Na^+ and H_2O following.

235. A tsunami tidal wave hits the east coast of South America and the people living there are forced to drink unclean water. Within the next several days, a large number of people develop severe diarrhea and about half of these people expire. Samples of drinking water are positive for *Vibrio cholerae*. Which of the following types of ion channels is most likely to be irreversibly opened in the epithelial cells of the crypts of Lieberkuhn in these people with severe diarrhea?

(A) Calcium channels
(B) Chloride channels
(C) Magnesium channels
(D) Potassium channels
(E) Sodium channels

Answer: B. Cholera toxin causes an irreversible increase in cAMP levels in enterocytes, which leads to an irreversible opening of chloride channels on the luminal surface. Movement of chloride into the gut lumen causes a secondary movement of sodium ions through paracellular pathways into the gut lumen. Water follows the osmotic gradient causing a tremendous increase in fluid loss into the gut lumen, which results in severe diarrhea.

236. After a long workout, a third-year medical student drinks a bottle of an electrolyte containing sports drink. Which of the following is the major mechanism for absorption of sodium from the small intestine?

(A) Na^+-H^+ exchange
(B) Cotransport with potassium
(C) Electrogenic transport
(D) Neutral NaCl absorption
(E) Solvent drag

Answer: D. Although multiple pathways exist for the absorption of Na^+, neutral absorption is the major mechanism. Absorption of sodium is the primary absorptive event in the small intestine. Absorption of Na^+ is necessary for absorption of water and other electrolytes. Neutral absorption may occur in two ways: Na^+ cotransported with Cl^- or in exchange for H^+ ions.

237. A 27-year-old woman comes to the emergency room because of a 2-day bout of profuse watery diarrhea. Physical examination reveals dry lips and oropharynx. The patient is diagnosed with acute secretory diarrhea and dehydration, likely due to Escherichia coli. Which of the following sodium reabsorptive pathways is inhibited by the enterotoxin?

(A) Sodium-glucose-coupled cotransport
(B) Electroneutral NaCl transport
(C) Sodium-phosphorous countertransport
(D) Sodium-hydrogen countertransport
(E) Sodium-bile salt cotransport

Answer: B. Diarrhea is the abnormal passage of fluid or semisolid stool with increased frequency. The general approach to the primary evaluation of patients with diarrhea is differentiating between an infectious versus noninfectious etiology. Although sodium is absorbed from the small intestine by several mechanisms, bacterial toxins specifically inhibit neutral NaCl absorption. In addition, the toxins augment diarrhea by increasing salt and water secretion by intestinal crypt cells. Oral rehydration involves utilizing the sodium-glucose-coupled cotransport pathway

238. A 14-year-old ballerina reports that she has chronic diarrhea. A detailed history reveals that she frequently drinks skim milk, that she does not use laxatives, and that she has noticed that her condition improves during times that she fasts for religious observances. In contrast to secretory diarrhea, which of the following is most likely seen with osmotic diarrhea?

(A) It is characterized by an increase in the stool osmotic gap.
(B) It is the result of increased crypt cell secretion.
(C) It is the result of decreased electroneutral sodium absorption.
(D) It is caused by bacterial toxins.
(E) It occurs only in the colon.

Answer: A. Osmotic diarrhea occurs when ingested, poorly absorbable, osmotically active solutes draw fluid into the lumen of the small intestine or colon leading to osmotic water loss in the stool. In osmotic diarrhea, the stool osmotic gap ($290 - 2[Na^+] + [K^+]$) exceeds 50 mOsm, consistent with an unmeasured solute contributing to the fecal electrolyte content. Osmotic diarrhea generally ceases with fasting or discontinued ingestion of the solute. The most common causes of osmotic diarrhea are (1) lactase (and other disaccharide) deficiency with resultant lactose intolerance and carbohydrate malabsorption, (2) ingestion of magnesium-containing antacids or laxatives, and (3) ingestion of non-absorbable sugars, such as sorbitol.

Secretory diarrhea, on the other hand, is caused by the overproduction of water by the small and large bowel. In contrast to osmotic diarrhea, secretory diarrhea has a normal stool osmotic gap and is not remedied with fasting. The other major pathophysiologic mechanisms of chronic diarrhea include steatorrheal, inflammatory infectious, dysmotile, radiation injury and factitial causes.

239. A 52-year-old patient with past medical history significant for cirrhosis and recent total colectomy presents to the ER with hepatic encephalopathy. Removal of the entire colon would be expected to cause which of the following?

(A) Decreased urinary urobilinogen
(B) An increase in blood ammonia
(C) Megaloblastic anemia
(D) Severe malnutrition
(E) Death

Answer: B. Most of the urea is formed in the liver, and in severe liver disease the blood urea nitrogen falls and blood NH_3 rises. One of the actions of colonic bacteria is to convert NH_3 to NH_4^+. Thus, a total colectomy would increase blood ammonia levels, which would be exacerbated in a person with cirrhosis. NH_3 is in equilibrium with NH_4^+. Humans can survive after total removal of the colon if fluid and electrolyte balance is maintained. When total colectomy is performed, the ileum is brought out through the abdominal wall (ileostomy). Advances in surgical techniques make ileostomies relatively trouble-free and patients with them can lead essentially normal lives. In liver disease (such as hepatitis), the intrahepatic urobilinogen cycle is inhibited also increasing urobilinogen levels.

240. A 27-year-old female patient with a history of irritable bowel syndrome presents with a chief complaint of flatulence. Gas within the colon is primarily derived from which of the following sources?

(A) CO_2 liberated by the interaction of HCO_3^- and H^+
(B) Diffusion from the blood
(C) Fermentation of undigested oligosaccharides by bacteria
(D) Swallowed atmospheric air
(E) Air pockets within foodstuffs

Answer: C. Gas within the colon is derived primarily from fermentation of undigested material by intestinal bacteria to produce CO_2, H_2, and methane. The digestive tract normally contains about 150 to 200 mL of gas, most of which is in the colon (100-150 mL). Most of the gas in the stomach is derived from air swallowed during eating or in periods of anxiety. Gas is produced in the small intestine by interaction of gastric acid and bicarbonate in the intestinal and pancreatic secretions but does not accumulate because it is either reabsorbed or quickly passed into the colon. The amount of gas varies markedly from one person to another and is influenced by diet; for example, ingestion of large amounts of beans, which contain indigestible carbohydrates in their hulls, will increase gas formation by intestinal bacteria. Diffusion of gas from the blood to the intestinal lumen is responsible for the N_2 present in intestinal gas and is influenced by the atmospheric pressure. While many patients with irritable bowel syndrome complain of increased flatus, most produce no more than that seen in normal individuals.

Colon and K^+

241. Potassium is normally:

(A) Passively absorbed in the small intestine and actively secreted in the proximal colon
(B) Passively absorbed in the small intestine and actively absorbed in the proximal colon
(C) Actively absorbed throughout small and large intestine
(D) Passively absorbed throughout small and large intestine
(E) Actively secreted in the small intestine and actively absorbed in the proximal colon

Answer: A. Throughout the small intestine there is a net reabsorption of potassium. However, in the colon there is net secretion, and because of this, diarrhea can lead to hypokalemia.

242. Where does the secretion of K^+ leads to excretion?

(A) Fundus of the stomach
(B) Antrum of the stomach
(C) Duodenum of the intestine
(D) Ileum of the intestine
(E) Colon

Answer: E. The colon secretes K^+ into the chyme. About 10% of the daily K^+ load is excreted by the colon; the remainder is excreted by the kidney.

243. A 37-year-old man presents with dehydration and hypokalemic metabolic acidosis. This acid-base and electrolyte disorder can occur with excess fluid loss from which of the following organs?

(A) Stomach
(B) Ileum
(C) Colon
(D) Pancreas
(E) Liver

Answer: C. Excessive loss of fluid from the gastrointestinal tract can lead to dehydration and, depending on the origin of the fluid loss, electrolyte and acid-base disturbances. $[H^+]$ and $[K^+]$ of gastric juice exceeds that of the plasma. As a result, excess fluid loss leads to metabolic alkalosis accompanied by hypokalemia. Because the pancreas, liver, ileum, and colon secrete bicarbonate, excessive loss leads to metabolic acidosis. In addition, the colon secretes potassium. Thus, the acidosis is accompanied by hypokalemia.

244. A patient comes into your office complaining of constant diarrhea after eating a 3 week old sandwich, she found underneath her bed. You are worried about electrolyte levels and take a blood sample. Which one of the following would best describe K^+, HCO_3^- and H^+ levels?

	K^+	HCO_3^-	H^+
(A)	Low	High	Low
(B)	High	High	High
(C)	High	Low	High
(D)	Low	Low	High
(E)	High	High	Low

Answer: D. In diarrhea, direct loss of bicarbonate leads to acidosis. Potassium is lost directly with GI secretion.